

Econometrics in R

Lecture 4

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M1 APE - Fall 2022



What we've seen so far

Manipulate data with `dplyr`

```
read.csv("ligue1.csv")
```

```
#  
#  
#  
#  
#  
#
```

##	Wk	Day	Date	Time	Home	xG	Score	xG.1	Away	Attendance	...
## 1	1	Fri	2021-08-06	21:00	Monaco	2.0	1-1	0.3	Nantes	7500	...
## 2	1	Sat	2021-08-07	17:00	Lyon	1.4	1-1	0.8	Brest	29018	...
## 3	1	Sat	2021-08-07	21:00	Troyes	0.8	1-2	1.2	Paris S-G	15248	...
## 4	1	Sun	2021-08-08	13:00	Rennes	0.6	1-1	2.0	Lens	22567	...
## 5	1	Sun	2021-08-08	15:00	Bordeaux	0.7	0-2	3.3	Clermont Foot	18748	...
## 6	1	Sun	2021-08-08	15:00	Strasbourg	0.4	0-2	0.9	Angers	23250	...
## 7	1	Sun	2021-08-08	15:00	Nice	0.8	0-0	0.2	Reims	18030	...
## 8	1	Sun	2021-08-08	15:00	Saint-Étienne	2.1	1-1	1.3	Lorient	20461	...
## 9	1	Sun	2021-08-08	17:00	Metz	0.7	3-3	1.4	Lille	15551	...
...	...	...	...	...	...	...	...	...	...	...	...



What we've seen so far

Manipulate data with `dplyr`

```
read.csv("ligue1.csv") %>%  
  select(Home, xG, Score, xG.1, Away)           # Keep/drop certain columns  
#  
#  
#  
#  
#  
#
```

```
##           Home  xG  Score  xG.1      Away  
## 1      Monaco 2.0   1-1   0.3      Nantes  
## 2        Lyon 1.4   1-1   0.8        Brest  
## 3      Troyes 0.8   1-2   1.2    Paris S-G  
## 4      Rennes 0.6   1-1   2.0        Lens  
## 5    Bordeaux 0.7   0-2   3.3 Clermont Foot  
## 6   Strasbourg 0.4   0-2   0.9        Angers  
## 7         Nice 0.8   0-0   0.2        Reims  
## 8 Saint-Étienne 2.1   1-1   1.3      Lorient  
## 9         Metz 0.7   3-3   1.4        Lille  
## ...           ...  ...   ...   ...           ...
```



What we've seen so far

Manipulate data with `dplyr`

```
read.csv("ligue1.csv") %>%  
  select(Home, xG, Score, xG.1, Away) %>%  
  mutate(home_winner = xG > xG.1)
```

Keep/drop certain columns
Create a new variable
#
#
#
#

	Home	xG	Score	xG.1	Away	home_winner
## 1	Monaco	2.0	1-1	0.3	Nantes	TRUE
## 2	Lyon	1.4	1-1	0.8	Brest	TRUE
## 3	Troyes	0.8	1-2	1.2	Paris S-G	FALSE
## 4	Rennes	0.6	1-1	2.0	Lens	FALSE
## 5	Bordeaux	0.7	0-2	3.3	Clermont Foot	FALSE
## 6	Strasbourg	0.4	0-2	0.9	Angers	FALSE
## 7	Nice	0.8	0-0	0.2	Reims	TRUE
## 8	Saint-Étienne	2.1	1-1	1.3	Lorient	TRUE
## 9	Metz	0.7	3-3	1.4	Lille	FALSE
...	...	...	...	...	...	...



What we've seen so far

Manipulate data with `dplyr`

```
read.csv("ligue1.csv") %>%  
  select(Home, xG, Score, xG.1, Away) %>%  
  mutate(home_winner = xG > xG.1) %>%  
  filter(Home == "Rennes")
```

Keep/drop certain columns
Create a new variable
Keep/drop certain rows
#
#
#

##		Home	xG	Score	xG.1	Away	home_winner
## 1		Rennes	0.6	1-1	2.0	Lens	FALSE
## 2		Rennes	0.9	1-0	0.5	Nantes	TRUE
## 3		Rennes	1.0	0-2	0.5	Reims	TRUE
## 4		Rennes	2.4	6-0	0.3	Clermont Foot	TRUE
## 5		Rennes	0.8	2-0	1.4	Paris S-G	FALSE
## 6		Rennes	1.5	1-0	0.6	Strasbourg	TRUE
## 7		Rennes	3.8	4-1	1.1	Lyon	TRUE
## 8		Rennes	3.1	2-0	0.7	Montpellier	TRUE
## 9		Rennes	0.8	1-2	0.6	Lille	TRUE
		...	...	...	...	...	...



What we've seen so far

Manipulate data with `dplyr`

```
read.csv("ligue1.csv") %>%  
  select(Home, xG, Score, xG.1, Away) %>%  
  mutate(home_winner = xG > xG.1) %>%  
  filter(Home == "Rennes") %>%  
  arrange(-xG)  
  
# Keep/drop certain columns  
# Create a new variable  
# Keep/drop certain rows  
# Sort rows  
#  
#
```

##		Home	xG	Score	xG.1	Away	home_winner
## 1		Rennes	3.8	4-1	1.1	Lyon	TRUE
## 2		Rennes	3.3	6-0	0.4	Bordeaux	TRUE
## 3		Rennes	3.3	6-1	0.9	Metz	TRUE
## 4		Rennes	3.1	2-0	0.7	Montpellier	TRUE
## 5		Rennes	2.7	2-0	0.3	Brest	TRUE
## 6		Rennes	2.6	4-1	0.4	Troyes	TRUE
## 7		Rennes	2.4	6-0	0.3	Clermont Foot	TRUE
## 8		Rennes	1.9	2-3	2.9	Monaco	FALSE
## 9		Rennes	1.7	2-0	0.3	Angers	TRUE
		...	...	...	...	...	...



What we've seen so far

Manipulate data with `dplyr`

```
read.csv("ligue1.csv") %>%  
  select(Home, xG, Score, xG.1, Away) %>%  
  mutate(home_winner = xG > xG.1) %>%  
  filter(Home == "Rennes") %>%  
  arrange(-xG) %>%  
  summarise(expected_wins = mean(home_winner),  
            expected_goals = sum(xG))
```

Keep/drop certain columns
Create a new variable
Keep/drop certain rows
Sort rows
Aggregate into statistics
#

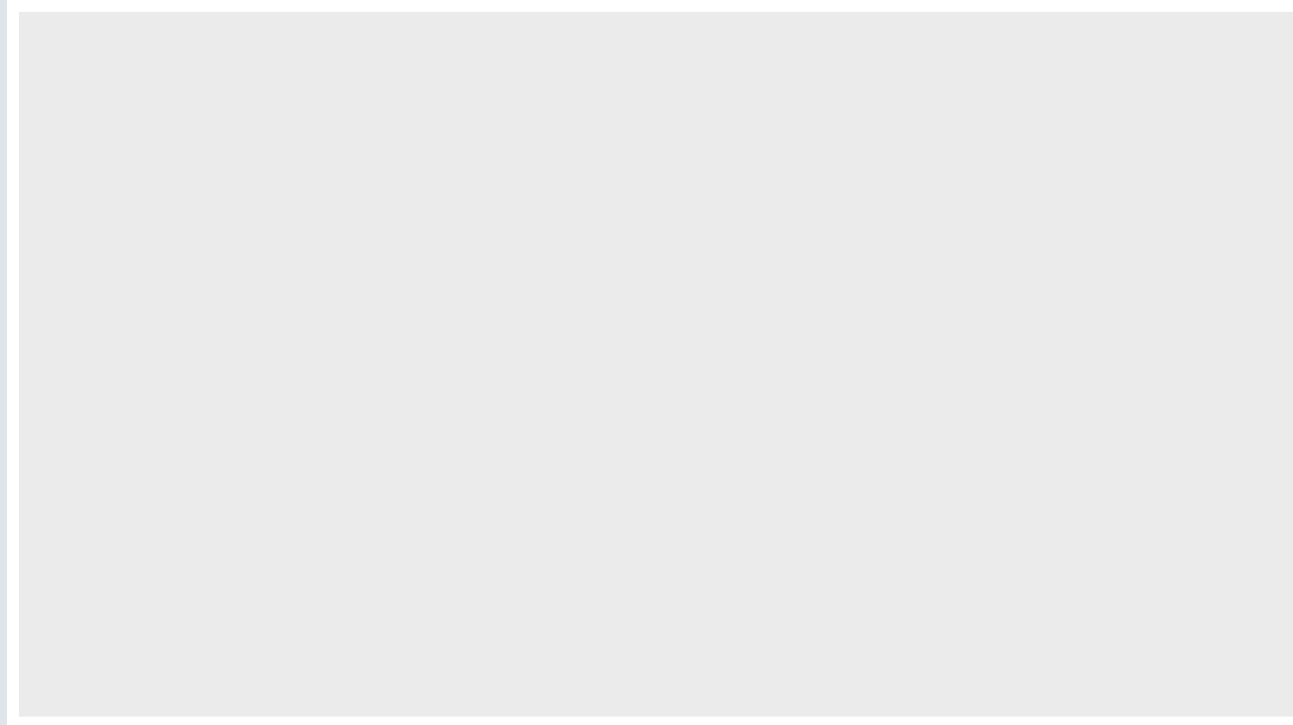
```
##      expected_wins expected_goals  
## 1          0.8421053          36.6
```



What we've seen so far

Plot data with `ggplot()`

```
ggplot(read.csv("wid.csv"))           # Data  
                                     #
```

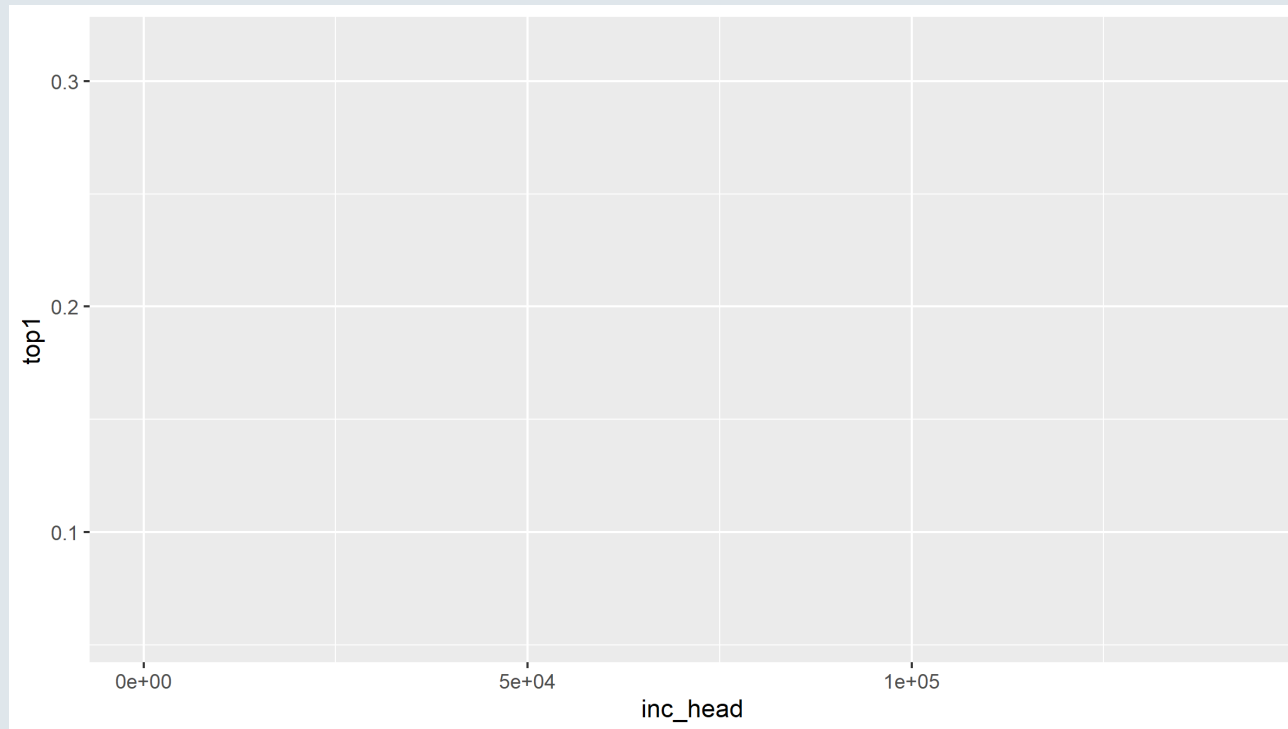




What we've seen so far

Plot data with `ggplot()`

```
ggplot(read.csv("wid.csv"), aes(x = inc_head, y = top1))      # Data & aesthetics  
#
```



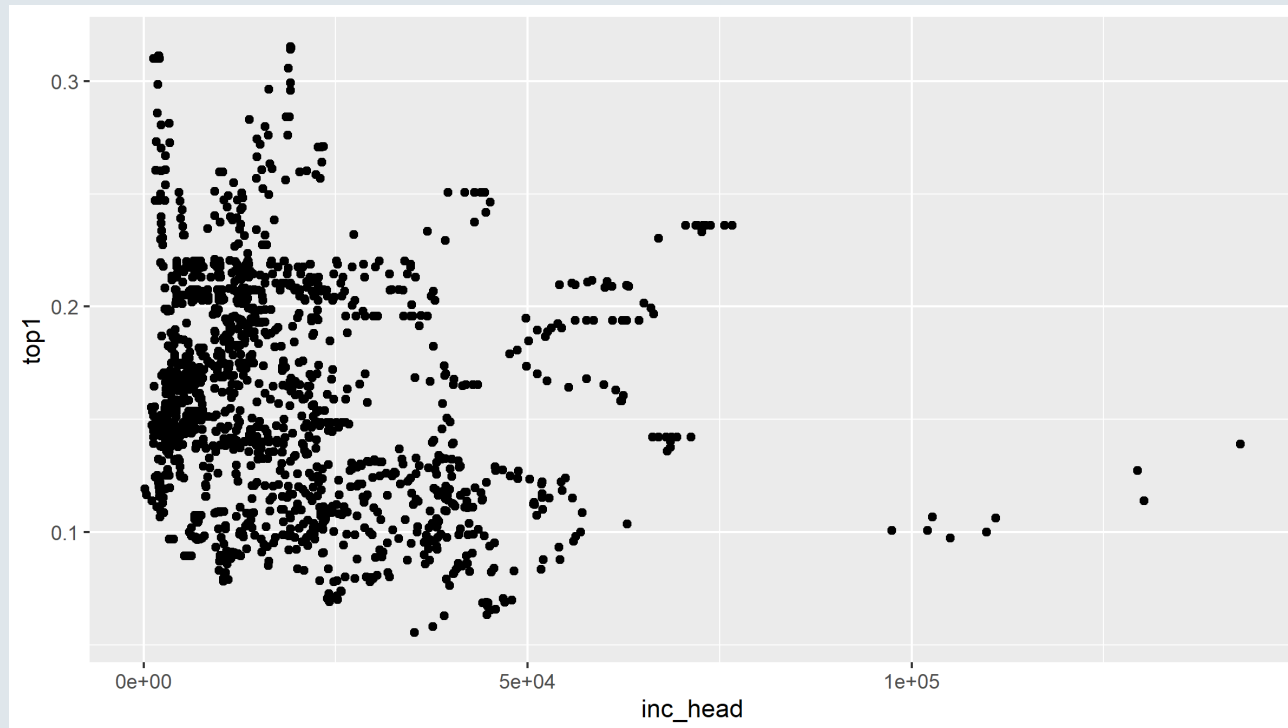


What we've seen so far

Plot data with `ggplot()`

```
ggplot(read.csv("wid.csv"), aes(x = inc_head, y = top1)) +  
  geom_point()
```

Data & aesthetics
Geometry





What we've seen so far

Write reports with R markdown

```
---  
title: "Starbucks"  
author: "Louis Sirugue"  
output: html_document  
---
```

Starbucks

Louis Sirugue



What we've seen so far

Write reports with R markdown

```
---  
title: "Starbucks"  
author: "Louis Sirugue"  
output: html_document  
---  
  
```${r, echo = F, message = F, warning = F}  
library(ggplot2) # Load package
starbucks <- read.csv("starbucks.csv", sep = ";") # Load data
```  
  
Nutritional values of `r nrow(starbucks)` *starbucks* beverages
```

Starbucks

Louis Sirugue

Nutritional values of 242 *starbucks* beverages



What we've seen so far

Write reports with R markdown

```
---
title: "Starbucks"
author: "Louis Sirugue"
output: html_document
---

```{r, echo = F, message = F, warning = F}
library(ggplot2) # Load package
starbucks <- read.csv("starbucks.csv", sep = ";") # Load data
```

Nutritional values of `r nrow(starbucks)` starbucks beverages

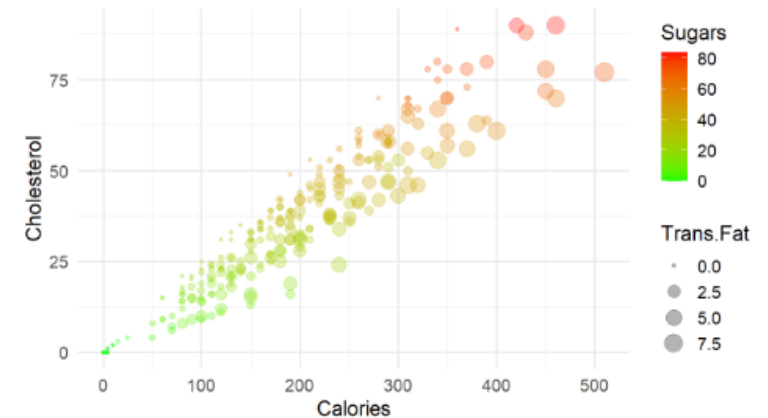
```{r, fig.height = 4}
ggplot(starbucks, aes(x = Calories, y = Cholesterol,
 size = Trans.Fat, color = Sugars)) +
 geom_point(alpha = .3) + theme_minimal(base_size = 14) +
 scale_color_gradient(low = "green", high = "red")
```
```

Starbucks

Louis Sirugue

Nutritional values of 242 *starbucks* beverages

```
ggplot(starbucks, aes(x = Calories, y = Cholesterol,
                      size = Trans.Fat, color = Sugars)) +
  geom_point(alpha = .3) + theme_minimal(base_size = 14) +
  scale_color_gradient(low = "green", high = "red")
```





Today: Econometrics in R!

1. Regressions

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables

2. Case study

- 2.1. Variable transformation
- 2.2. Functional form
- 2.3. Control variables
- 2.4. Interactions

3. Inference

- 3.1. Hypothesis testing
- 3.2. Confidence intervals

4. Report and export results

- 4.1. Regression tables
- 4.2. Plot coefficients

5. Wrap up!



Today: Econometrics in R!

1. Regressions

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables



1. Regressions

1.1. On continuous variables

- For this part we're going to work with the '**Great Gatsby Curve**'
 - It refers to the positive relationship between **inequality** and **intergenerational income persistence**
 - The term was coined by Alan Krueger based on the research of Miles Corak

```
ggcurve <- read.csv("ggcurve.csv")
str(ggcurve)
```

```
## 'data.frame':    22 obs. of  3 variables:
## $ country: chr  "Denmark" "Norway" "Finland" "Canada" ...
## $ ige : num  0.15 0.17 0.18 0.19 0.26 0.27 0.29 0.32 0.34 0.4 ...
## $ gini : num  0.378 0.325 0.378 0.463 0.439 ...
```

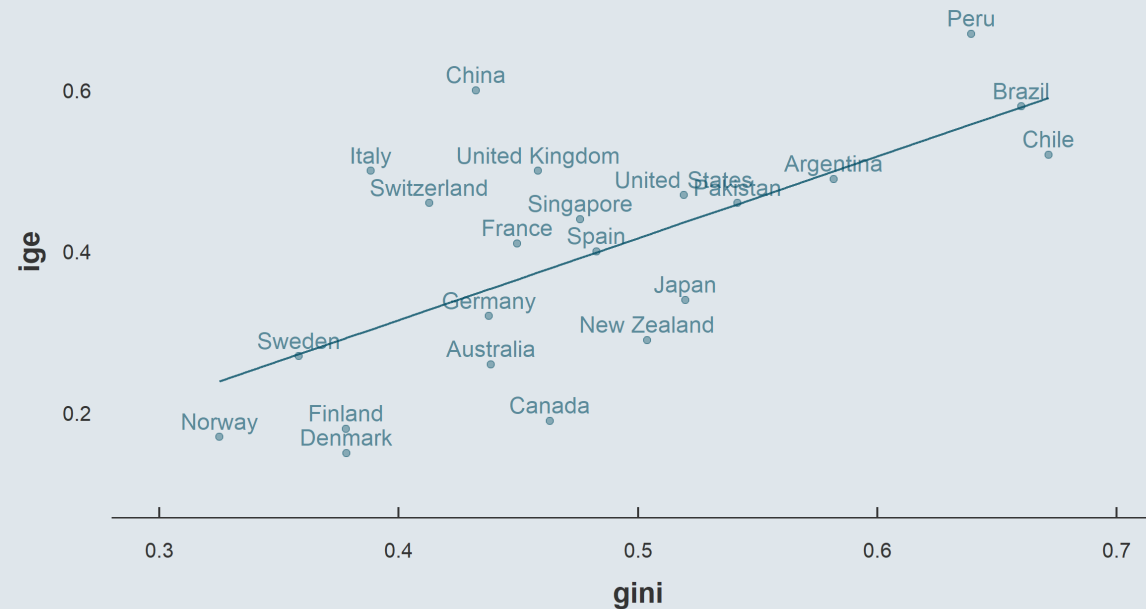
- For **22 countries** we have the following variables
 - **ige**: The intergenerational income elasticity, the higher the closer child income to parent income
 - **gini**: The Gini coefficient of income inequality, the higher the more concentrated the income



1. Regressions

1.1. On continuous variables

- You must already be quite familiar with univariate **regressions** $y = \alpha + \beta x + \varepsilon$

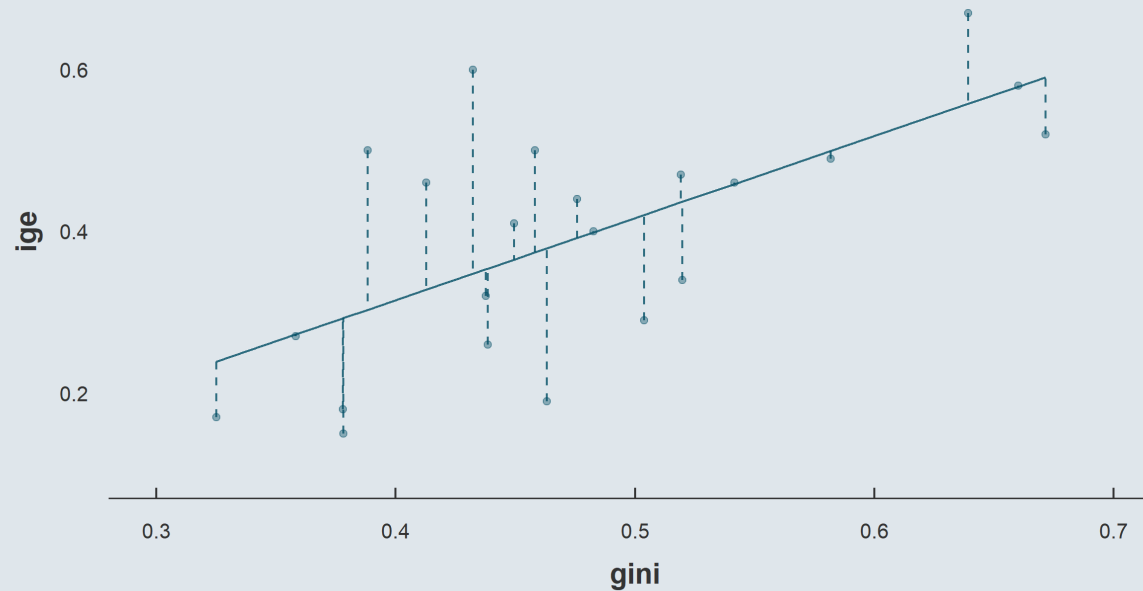




1. Regressions

1.1. On continuous variables

- You must already be quite familiar with univariate **regressions** $y = \alpha + \beta x + \varepsilon$
 - We're looking for the line $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ that **mimimizes distance** to the data points $\sum \varepsilon_i^2$

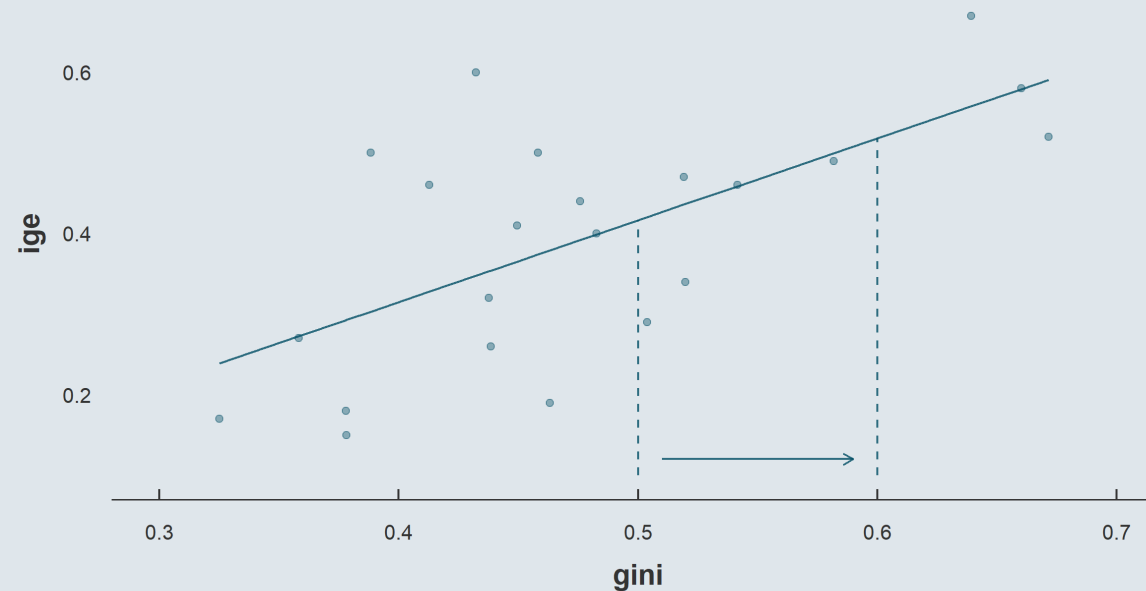




1. Regressions

1.1. On continuous variables

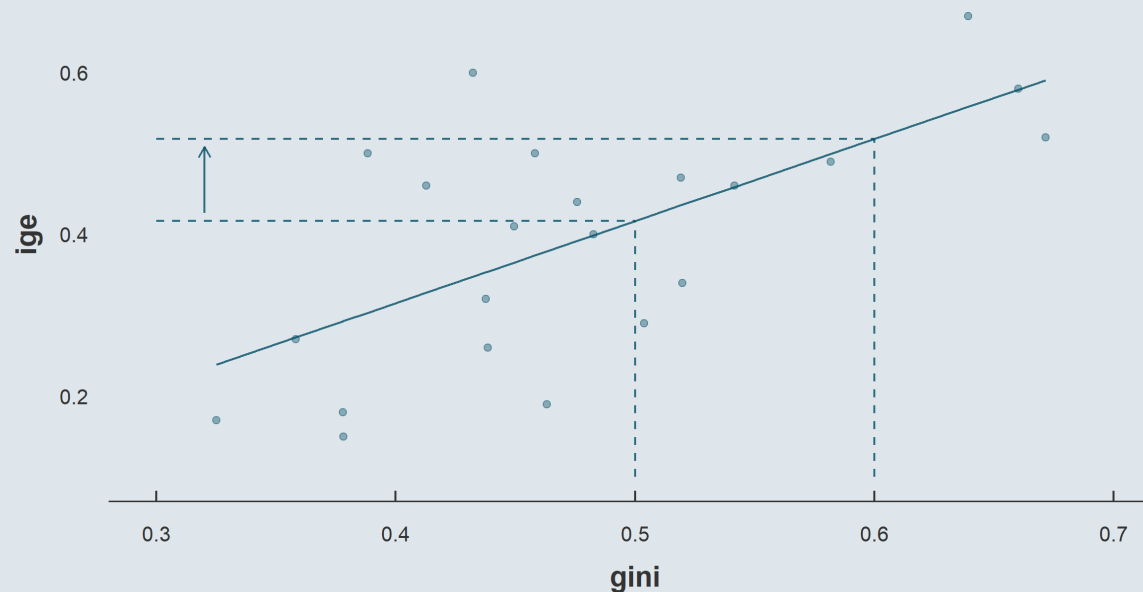
- You must already be quite familiar with univariate **regressions** $y = \alpha + \beta x + \varepsilon$
 - We're looking for the line $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ that **mimimizes distance** to the data points $\sum \varepsilon_i^2$
 - Such that for a **one unit increase** in x



1. Regressions

1.1. On continuous variables

- You must already be quite familiar with univariate **regressions** $y = \alpha + \beta x + \varepsilon$
 - We're looking for the line $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ that **mimimizes distance** to the data points $\sum \varepsilon_i^2$
 - Such that for a **one unit increase** in x
 - Its slope $\hat{\beta}$ indicates the associated **expected change** in y





1. Regressions

1.1. On continuous variables

- In R we can **estimate a regression** model using the **lm()** command (*for **Linear Model***)
 - 1.
 - 2.

```
lm( , )
```



1. Regressions

1.1. On continuous variables

- In R we can **estimate a regression** model using the **lm()** command (*for **Linear Model***)
 1. The first argument is the **formula**, written as **y ~ x**
 - 2.

```
lm(ige ~ gini, )
```

1. Regressions

1.1. On continuous variables

- In R we can **estimate a regression** model using the **lm()** command (*for **Linear Model***)
 1. The first argument is the **formula**, written as **y ~ x**
 2. The second argument is the **data** containing the variables

```
lm(ige ~ gini, ggcurve)
```

```
##  
## Call:  
## lm(formula = ige ~ gini, data = ggcurve)  
##  
## Coefficients:  
## (Intercept)          gini  
##    -0.09129      1.01546
```

- This is great but the **output** is a bit minimalistic
 - To get a more **exhaustive description** of our regression we can apply the **summary()** function to **lm()**

```
ggmodel <- lm(ige ~ gini, ggcurve) %>% summary()
```



1. Regressions

1.1. On continuous variables

```
ggmodel
```

```
##
## Call:
## lm(formula = ige ~ gini, data = ggcurve)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.188991 -0.088238 -0.000855  0.047284  0.252310
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09129    0.12870   -0.709   0.48631
## gini         1.01546    0.26425    3.843   0.00102 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1159 on 20 degrees of freedom
## Multiple R-squared:  0.4247,    Adjusted R-squared:  0.396
## F-statistic: 14.77 on 1 and 20 DF,  p-value: 0.001016
```



1. Regressions

1.1. On continuous variables

- It gives back the **command**

```
##  
## Call:  
## lm(formula = ige ~ gini, data = ggcurve)  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##
```

← Command



1. Regressions

1.1. On continuous variables

- A description of the **distribution** of **residuals**

```
##  
## Call:  
## lm(formula = ige ~ gini, data = ggcurve)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.188991 -0.088238 -0.000855  0.047284  0.252310  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##  
##
```

← Command

← Residuals distribution



1. Regressions

1.1. On continuous variables

- **Coefficients** along with their **standard error**, **t-value**, and **p-value**

```
##
## Call:
## lm(formula = ige ~ gini, data = ggcurve)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.188991 -0.088238 -0.000855  0.047284  0.252310
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09129     0.12870  -0.709   0.48631
## gini         1.01546     0.26425   3.843   0.00102
##
##
##
##
##
##
```

← Command

← Residuals distribution

← Coefs, s.e., t-/p-values



1. Regressions

1.1. On continuous variables

- **Significance** thresholds with symbols

```
##  
## Call:  
## lm(formula = ige ~ gini, data = ggcurve)
```

← Command

```
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max  
## -0.188991 -0.088238 -0.000855  0.047284  0.252310
```

← Residuals distribution

```
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept) -0.09129    0.12870  -0.709  0.48631  
## gini         1.01546    0.26425   3.843  0.00102 **
```

← Coefs, s.e., t-/p-values

```
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

← Significance

```
##  
##  
##  
##
```




1. Regressions

1.1. On continuous variables

- The **residual standard error** $\sqrt{\sum (y_i - \hat{y}_i)^2 / df}$ and **degrees of freedom**

```
##  
## Call:  
## lm(formula = ige ~ gini, data = ggcurve)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max  
## -0.188991 -0.088238 -0.000855  0.047284  0.252310  
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept) -0.09129     0.12870  -0.709   0.48631  
## gini         1.01546     0.26425   3.843   0.00102 **  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.1159 on 20 degrees of freedom  
##  
##
```

← Command

← Residuals distribution

← Coefs, s.e., t-/p-values

← Significance

← Residual s.e. & df.



1. Regressions

1.1. On continuous variables

- The R^2 and **adjusted R^2**

```
##  
## Call:  
## lm(formula = ige ~ gini, data = ggcurve)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.188991 -0.088238 -0.000855  0.047284  0.252310   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -0.09129     0.12870  -0.709   0.48631      
## gini         1.01546     0.26425   3.843   0.00102 **    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.1159 on 20 degrees of freedom  
## Multiple R-squared:  0.4247,    Adjusted R-squared:  0.396   
##
```

← Command

← Residuals distribution

← Coefs, s.e., t-/p-values

← Significance

← Residual s.e. & df.

← R^2 & adjusted R^2



1. Regressions

1.1. On continuous variables

- The result of an **F-test** ($H_0 : \beta_k = 0 \ \forall k$)

```
##
## Call:
## lm(formula = ige ~ gini, data = ggcurve)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.188991 -0.088238 -0.000855  0.047284  0.252310
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09129     0.12870  -0.709   0.48631
## gini         1.01546     0.26425   3.843   0.00102 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1159 on 20 degrees of freedom
## Multiple R-squared:  0.4247,    Adjusted R-squared:  0.396
## F-statistic: 14.77 on 1 and 20 DF,  p-value: 0.001016
```

← Command

← Residuals distribution

← Coefs, s.e., t-/p-values

← Significance

← Residual s.e. & df.

← R^2 & adjusted R^2

← F-test results



1. Regressions

1.1. On continuous variables

- All these elements are then easily accessible using the `$` operator

```
str(ggmodel, give.attr = F)
```

```
## List of 11
## $ call      : language lm(formula = ige ~ gini, data = ggcurve)
## $ terms     :Classes 'terms', 'formula' language ige ~ gini
## $ residuals : Named num [1:22] -0.1427 -0.0687 -0.1125 -0.189 -0.094 ...
## $ coefficients : num [1:2, 1:4] -0.0913 1.0155 0.1287 0.2642 -0.7093 ...
## $ aliases    : Named logi [1:2] FALSE FALSE
## $ sigma      : num 0.116
## $ df         : int [1:3] 2 20 2
## $ r.squared   : num 0.425
## $ adj.r.squared: num 0.396
## $ fstatistic  : Named num [1:3] 14.8 1 20
## $ cov.unscaled : num [1:2, 1:2] 1.23 -2.48 -2.48 5.19
```

→ Let's try it out!



1. Regressions

1.1. On continuous variables

- Take the **coefficients** for instance

```
ggmodel$coefficients
```

```
##              Estimate Std. Error   t value   Pr(>|t|)
## (Intercept) -0.09129311  0.1287045 -0.7093234 0.486311455
## gini         1.01546204  0.2642477  3.8428420 0.001015706
```

- We can **subset** this matrix like we would do with a regular `data.frame`

```
ggmodel$coefficients[2, 1]
```

```
## [1] 1.015462
```

```
ggmodel$coefficients[, "Pr(>|t|)"]
```

```
## (Intercept)      gini
## 0.486311455 0.001015706
```

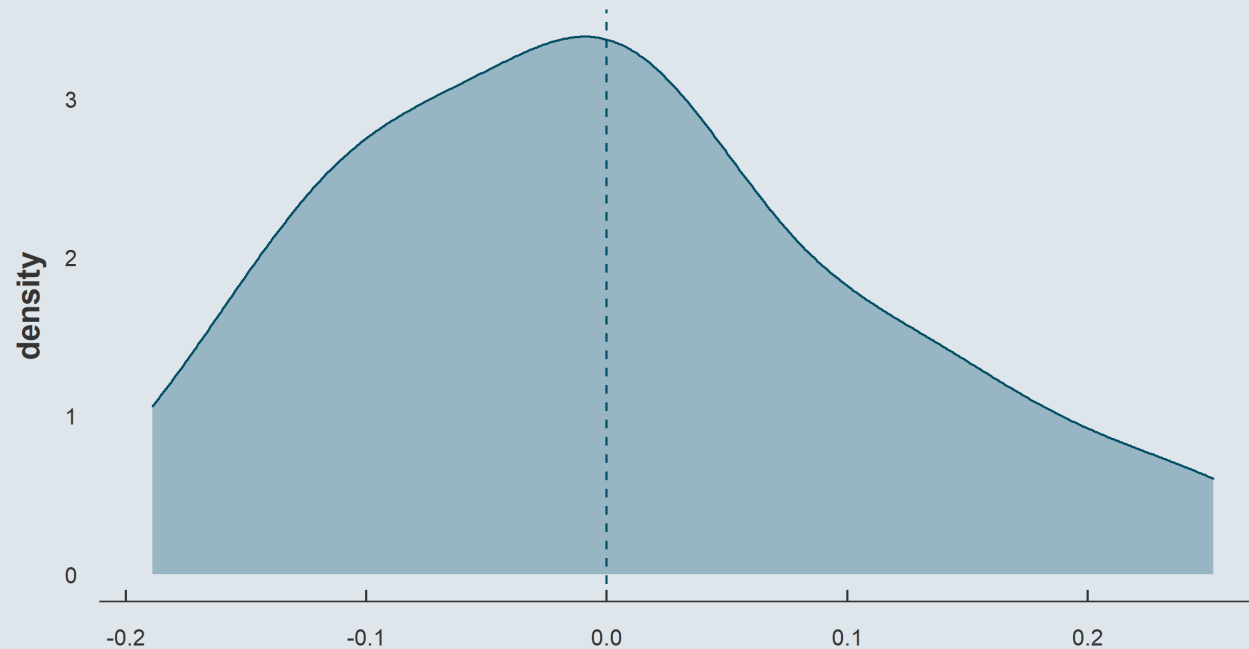


1. Regressions

1.1. On continuous variables

- We can also easily **plot** the **distribution** of our **residuals**

```
ggplot(data.frame(x = ggmodel$residuals), aes(x = x)) +  
  geom_density() + geom_vline(xintercept = 0, linetype = "dashed")
```



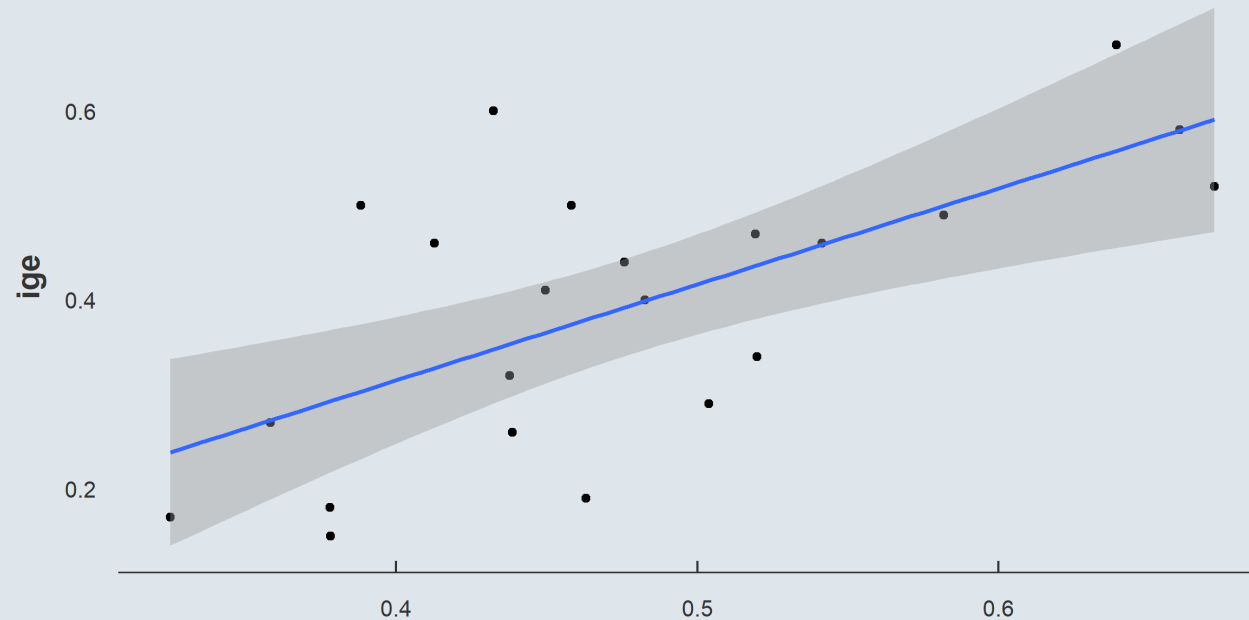


1. Regressions

1.1. On continuous variables

- Note that `ggplot()` has a dedicated **geometry for fitted values**: `geom_smooth()`

```
ggplot(ggcurve, aes(x = gini, y = ige)) +  
  geom_point() + geom_smooth(method = "lm")
```



Practice

10:00

Check that `lm()` works fine by computing the R^2 manually

1) Start by creating a variable for $\hat{\beta}$, then for $\hat{\alpha}$, $\hat{y}_i$, and $\hat{\varepsilon}_i$.

$$\hat{\beta} = \frac{\text{Cov}(x, y)}{\text{Var}(x)} \quad ; \quad \hat{\alpha} = \bar{y} - \hat{\beta} \times \bar{x}$$

You're gonna need the `cov()` and `var()` functions

2) Summarise the data into only $\hat{\alpha}$, $\hat{\beta}$, and the R^2

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

You've got 10 minutes!

Solution

```
ggcurve %>%  
  mutate(beta = cov(gini, ige) / var(gini))  
#  
#  
#  
#  
#  
#
```

```
##      country  ige      gini      beta  
## 1      Denmark 0.15 0.3781796 1.015462  
## 2       Norway 0.17 0.3250102 1.015462  
## 3      Finland 0.18 0.3779868 1.015462  
## 4       Canada 0.19 0.4631237 1.015462  
## 5    Australia 0.26 0.4385511 1.015462  
## 6       Sweden 0.27 0.3582480 1.015462  
## 7   New Zealand 0.29 0.5039373 1.015462  
## 8      Germany 0.32 0.4377010 1.015462  
## 9        Japan 0.34 0.5198383 1.015462  
## 10      Spain 0.40 0.4826243 1.015462  
## 11      France 0.41 0.4495654 1.015462  
## ..      ...    ...      ...      ...
```

Solution

```
ggcurve %>%  
  mutate(beta = cov(gini, ige) / var(gini),  
         alpha = mean(ige) - beta * mean(gini))  
#  
#  
#  
#  
#
```

```
##      country  ige      gini      beta      alpha  
## 1      Denmark 0.15 0.3781796 1.015462 -0.09129311  
## 2       Norway 0.17 0.3250102 1.015462 -0.09129311  
## 3      Finland 0.18 0.3779868 1.015462 -0.09129311  
## 4       Canada 0.19 0.4631237 1.015462 -0.09129311  
## 5    Australia 0.26 0.4385511 1.015462 -0.09129311  
## 6       Sweden 0.27 0.3582480 1.015462 -0.09129311  
## 7   New Zealand 0.29 0.5039373 1.015462 -0.09129311  
## 8      Germany 0.32 0.4377010 1.015462 -0.09129311  
## 9        Japan 0.34 0.5198383 1.015462 -0.09129311  
## 10      Spain 0.40 0.4826243 1.015462 -0.09129311  
## 11      France 0.41 0.4495654 1.015462 -0.09129311  
## ..      ...    ...      ...      ...      ...
```

Solution

```
ggcurve %>%  
  mutate(beta = cov(gini, ige) / var(gini),  
         alpha = mean(ige) - beta * mean(gini),  
         fit = alpha + beta * gini)  
  
#  
#  
#  
#
```

```
##      country  ige      gini      beta      alpha      fit  
## 1      Denmark 0.15 0.3781796 1.015462 -0.09129311 0.2927339  
## 2      Norway 0.17 0.3250102 1.015462 -0.09129311 0.2387424  
## 3      Finland 0.18 0.3779868 1.015462 -0.09129311 0.2925381  
## 4      Canada 0.19 0.4631237 1.015462 -0.09129311 0.3789915  
## 5    Australia 0.26 0.4385511 1.015462 -0.09129311 0.3540389  
## 6      Sweden 0.27 0.3582480 1.015462 -0.09129311 0.2724942  
## 7    New Zealand 0.29 0.5039373 1.015462 -0.09129311 0.4204361  
## 8      Germany 0.32 0.4377010 1.015462 -0.09129311 0.3531757  
## 9       Japan 0.34 0.5198383 1.015462 -0.09129311 0.4365829  
## 10     Spain 0.40 0.4826243 1.015462 -0.09129311 0.3987935  
## 11     France 0.41 0.4495654 1.015462 -0.09129311 0.3652234  
## ..      ...      ...      ...      ...      ...
```

Solution

```
ggcurve %>%  
  mutate(beta = cov(gini, ige) / var(gini),  
         alpha = mean(ige) - beta * mean(gini),  
         fit = alpha + beta * gini,  
         residuals = ige - fit)  
#  
#  
#
```

| ## | country | ige | gini | beta | alpha | fit | residuals |
|-------|-------------|------|-----------|----------|-------------|-----------|---------------|
| ## 1 | Denmark | 0.15 | 0.3781796 | 1.015462 | -0.09129311 | 0.2927339 | -0.1427339244 |
| ## 2 | Norway | 0.17 | 0.3250102 | 1.015462 | -0.09129311 | 0.2387424 | -0.0687424324 |
| ## 3 | Finland | 0.18 | 0.3779868 | 1.015462 | -0.09129311 | 0.2925381 | -0.1125381050 |
| ## 4 | Canada | 0.19 | 0.4631237 | 1.015462 | -0.09129311 | 0.3789915 | -0.1889914561 |
| ## 5 | Australia | 0.26 | 0.4385511 | 1.015462 | -0.09129311 | 0.3540389 | -0.0940388831 |
| ## 6 | Sweden | 0.27 | 0.3582480 | 1.015462 | -0.09129311 | 0.2724942 | -0.0024941569 |
| ## 7 | New Zealand | 0.29 | 0.5039373 | 1.015462 | -0.09129311 | 0.4204361 | -0.1304360777 |
| ## 8 | Germany | 0.32 | 0.4377010 | 1.015462 | -0.09129311 | 0.3531757 | -0.0331756674 |
| ## 9 | Japan | 0.34 | 0.5198383 | 1.015462 | -0.09129311 | 0.4365829 | -0.0965829037 |
| ## 10 | Spain | 0.40 | 0.4826243 | 1.015462 | -0.09129311 | 0.3987935 | 0.0012065012 |
| ## 11 | France | 0.41 | 0.4495654 | 1.015462 | -0.09129311 | 0.3652234 | 0.0447765627 |
| ## .. | ... | ... | ... | ... | ... | ... | ... |

Solution

```
ggcurve %>%  
  mutate(beta = cov(gini, ige) / var(gini),  
         alpha = mean(ige) - beta * mean(gini),  
         fit = alpha + beta * gini,  
         residuals = ige - fit) %>%  
  summarise(alpha = alpha[1],  
            beta = beta[1],  
            r2 = 1 - sum(residuals^2)/sum((ige - mean(ige))^2))
```

```
##           alpha      beta      r2  
## 1 -0.09129311  1.015462  0.424749
```

```
ggmodel$coefficients[, "Estimate"]
```

```
## (Intercept)      gini  
## -0.09129311  1.01546204
```

```
ggmodel$r.squared
```

```
## [1] 0.424749
```



1. Regressions

1.2. On binary variables

- Now consider that we want to know the **relationship** between **not** being a **European country** and the **ige**

```
ggcurve$country
```

```
## [1] "Denmark"      "Norway"       "Finland"      "Canada"
## [5] "Australia"    "Sweden"       "New Zealand"  "Germany"
## [9] "Japan"        "Spain"        "France"       "Singapore"
## [13] "Pakistan"     "Switzerland"  "United States" "Argentina"
## [17] "Italy"        "United Kingdom" "Chile"        "Brazil"
## [21] "China"        "Peru"
```

```
europe <- c("Denmark", "Norway", "Finland", "Sweden", "Germany", "Spain",
           "France", "Switzerland", "Italy", "United Kingdom")
```

```
ggcurve <- ggcurve %>%
  mutate(continent = ifelse(country %in% europe, "Europe", "Other"))
```



1. Regressions

1.2. On binary variables

- Can we just **regress** **ige** **on** **continent** even though it's a **character** variable?

```
lm(ige ~ continent, ggcurve)
```

```
##  
## Call:  
## lm(formula = ige ~ continent, data = ggcurve)  
##  
## Coefficients:  
##      (Intercept)  continentOther  
##           0.3360           0.1065
```

- Seems like we can!

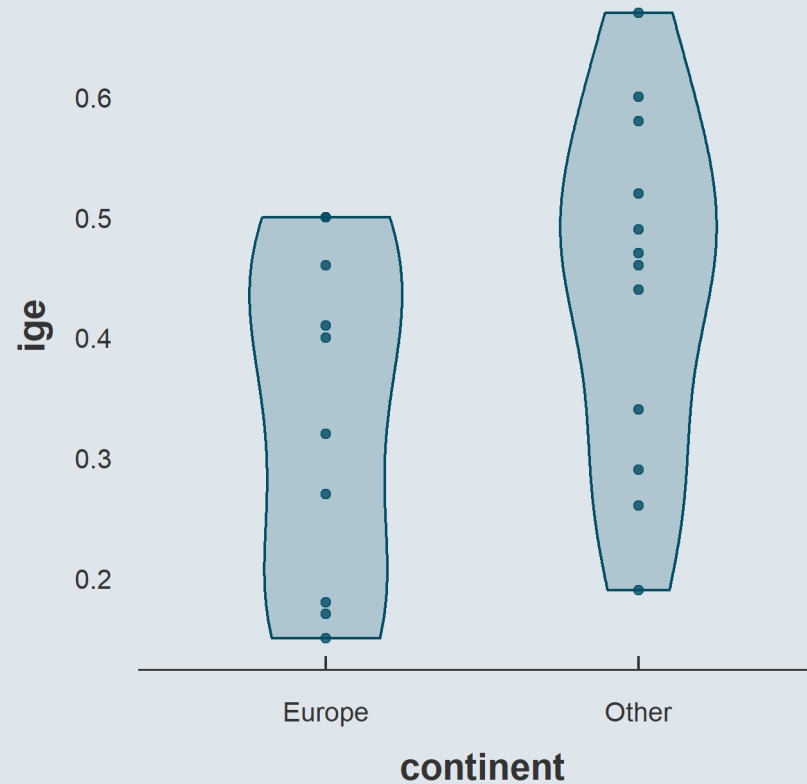
→ *But what's actually going on?*



1. Regressions

1.2. On binary variables

- R implicitly **converts** character variables into **binary** variables

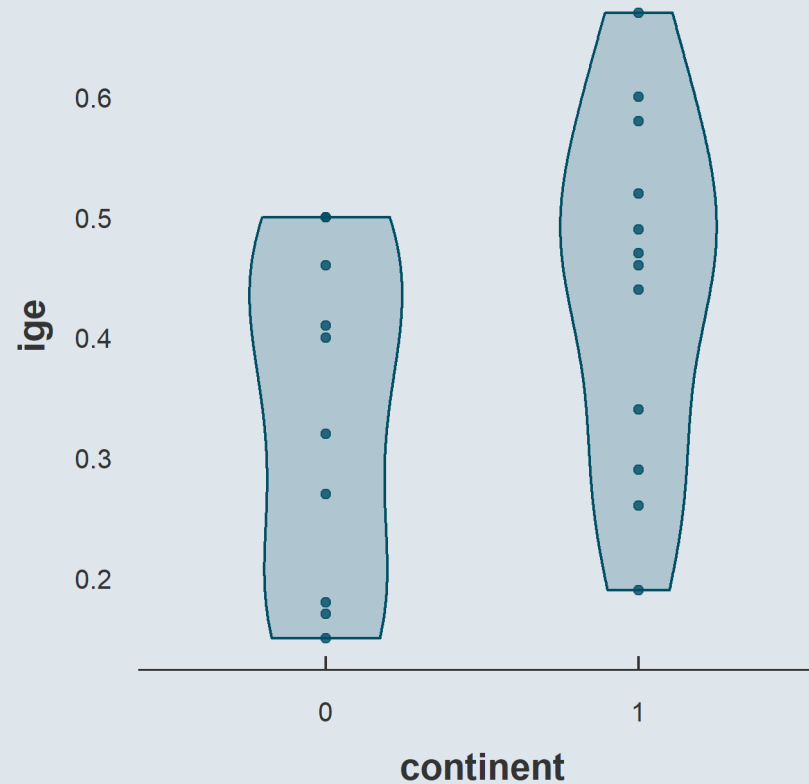




1. Regressions

1.2. On binary variables

- Such that just **like** in the **continuous** case...

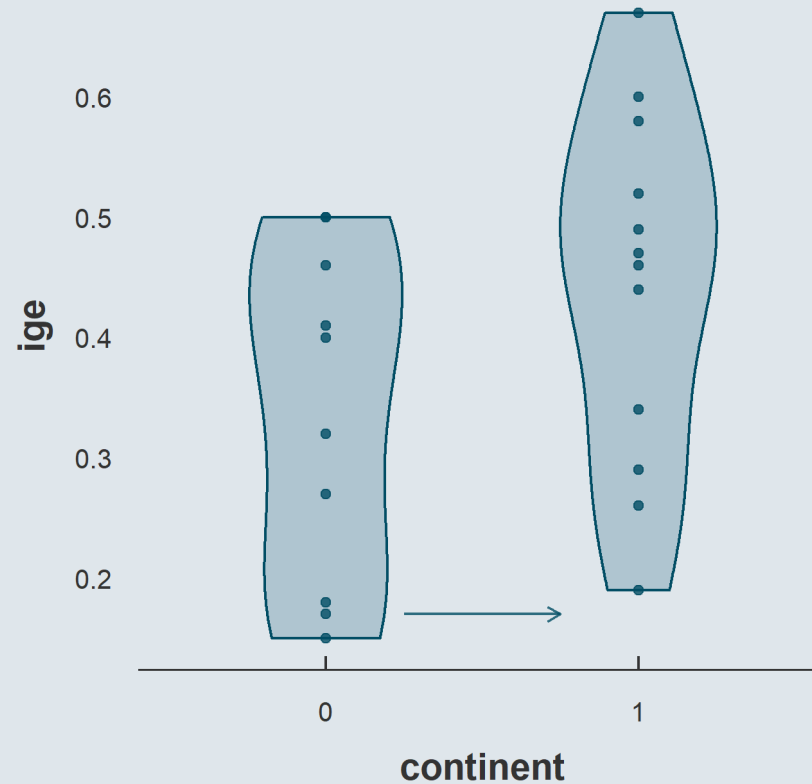




1. Regressions

1.2. On binary variables

- ... we're looking at a **1-unit increase** in x (i.e., switching from 0 (Europe) to 1 (Other))

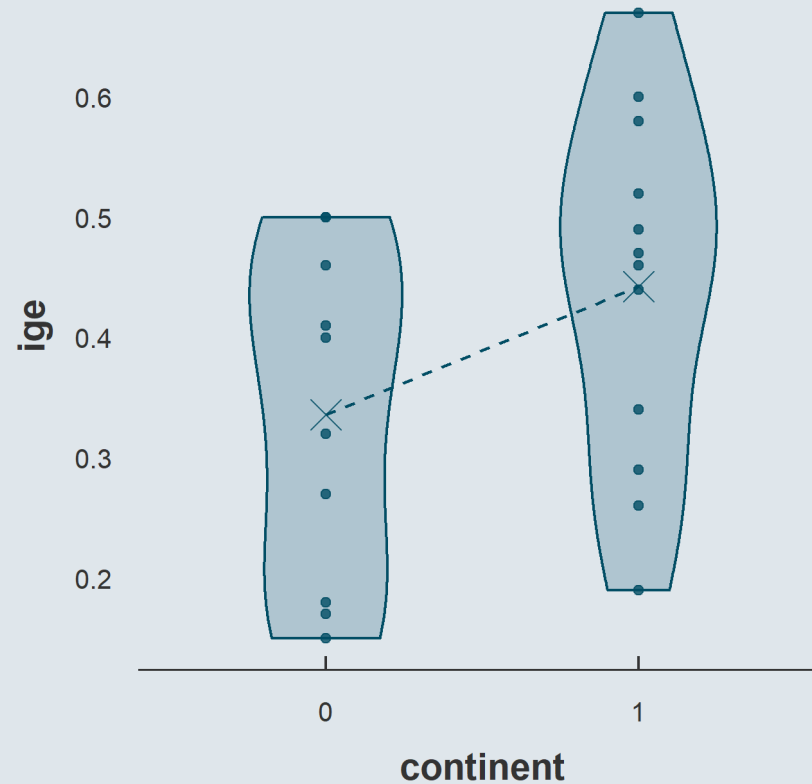




1. Regressions

1.2. On binary variables

- As you know the **fit** is necessarily going through the **mean of each category**

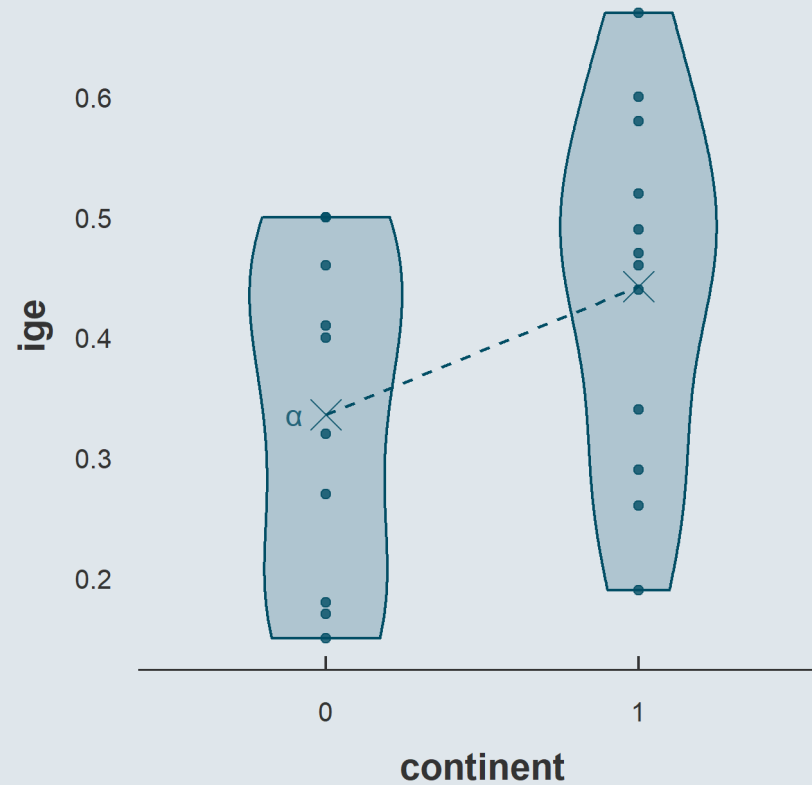




1. Regressions

1.2. On binary variables

- Such that $\hat{\alpha}$ is the **mean of the reference group**

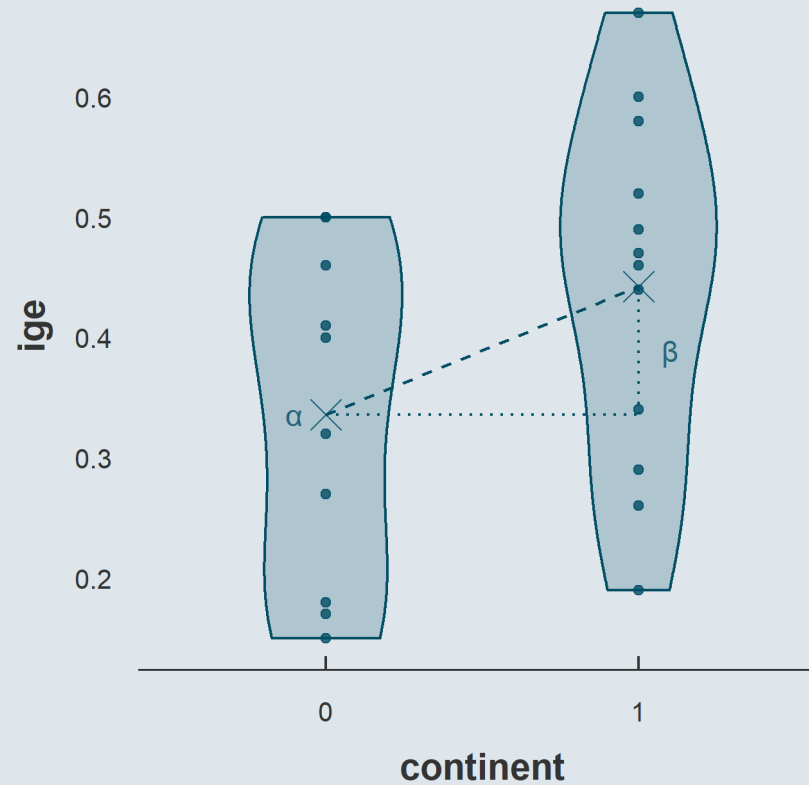




1. Regressions

1.2. On binary variables

- And $\hat{\beta}$ is the **difference in means**





1. Regressions

1.2. On binary variables

- We can verify that easily:

```
ggcurve %>%  
  group_by(continent) %>%  
  summarise(ybar = mean(ige)) %>%  
  mutate(relative = ybar - ybar[1])
```

```
## # A tibble: 2 x 3  
##   continent ybar relative  
##   <chr>     <dbl>   <dbl>  
## 1 Europe    0.336     0  
## 2 Other     0.442    0.106
```

```
lm(ige ~ continent, ggcurve)
```

```
##  
## Call:  
## lm(formula = ige ~ continent, data = ggcurve)  
##  
## Coefficients:  
##   (Intercept)  continentOther  
##          0.3360          0.1065
```

→ And what about discrete variables with more than 2 categories?



1. Regressions

1.3. On categorical variables

- Let's work on the [2020 Annual Social and Economic \(ASEC\) Supplement](#) to the US CPS
 - Here is an extract on 64,336 working individuals with positive earnings
 - For which I kept only 4 variables:

```
asec <- read.csv("asec.csv")
str(asec)
```

```
## 'data.frame':    64336 obs. of  4 variables:
## $ Sex      : chr  "Female" "Male" "Female" "Male" ...
## $ Earnings: int   52500 34000 40000 8424 58000 42000 55000 28000 200 25000 ...
## $ Race     : chr  "White" "White" "White" "White" ...
## $ Hours    : int   40 40 44 21 60 40 40 40 20 40 ...
```

- Let's say we want to regress earnings on Race

```
unique(asec$Race)
```

```
## [1] "White" "Other" "Black"
```



1. Regressions

1.3. On categorical variables

- Just like the **2-category** variable was equivalent to **1 dummy** variable
 - An **n-category** variable is equivalent to **n-1 dummy** variables

| Continent | Other | | Race | Other | White |
|-----------|-------|--|-------|-------|-------|
| Europe | 0 | | Black | 0 | 0 |
| Europe | 0 | | Black | 0 | 0 |
| Europe | 0 | | Other | 1 | 0 |
| Other | 1 | | Other | 1 | 0 |
| Other | 1 | | White | 0 | 1 |
| Other | 1 | | White | 0 | 1 |

```
lm(Earnings ~ Race, asec)
```

```
##  
## Call:  
## lm(formula = Earnings ~ Race, data = asec)  
##  
## Coefficients:  
## (Intercept)      RaceOther      RaceWhite  
##      50577      17477      12303
```

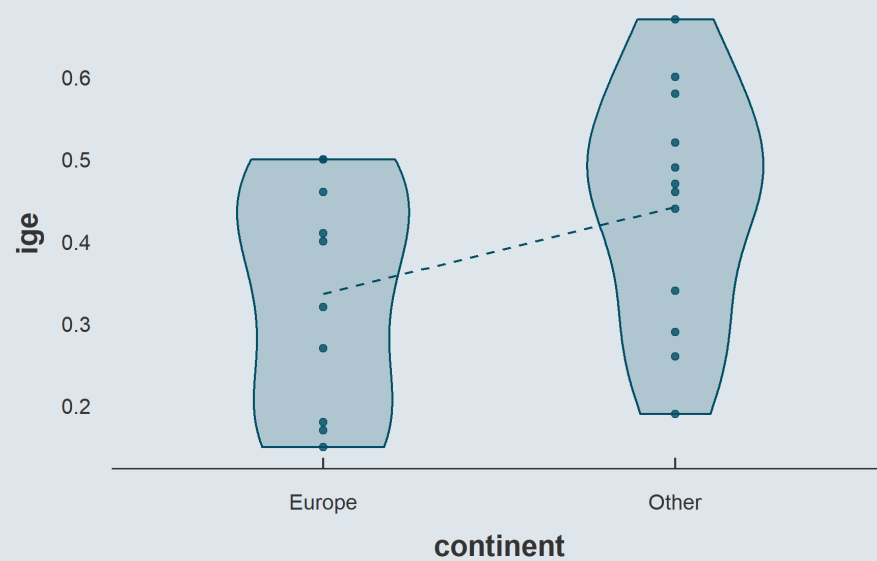
- Instead of 1 x axis we're going to have $n-1$ x axes



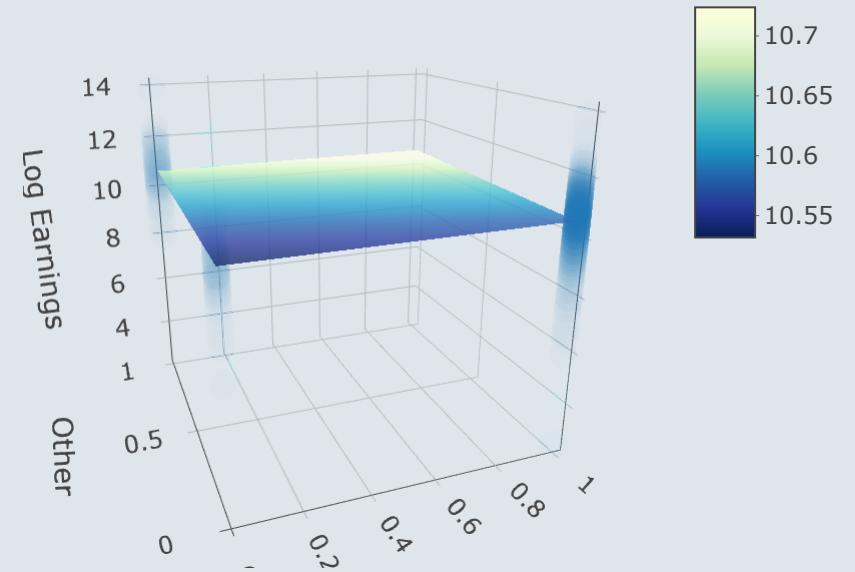
1. Regressions

1.3. On categorical variables

2-category variable



3-category variable





1. Regressions

1.3. On categorical variables

- Once again the **constant** is the **average** y and the **slopes** are the relative **differences in means**

```
lm(Earnings ~ Race, asec)
```

```
##  
## Call:  
## lm(formula = Earnings ~ Race, data = asec)  
##  
## Coefficients:  
## (Intercept)      RaceOther      RaceWhite  
##          50577          17477          12303
```

```
asec %>%  
  group_by(Race) %>%  
  summarise(ybar = mean(Earnings)) %>%  
  mutate(relative = ybar - ybar[1])
```

```
## # A tibble: 3 x 3  
##   Race      ybar relative  
##   <chr>   <dbl>   <dbl>  
## 1 Black  50577.      0  
## 2 Other  68055.  17477.  
## 3 White  62880.  12303.
```

- Note that R always sorts **character** variables by **alphabetical order**
 - But it would be more natural to interpret relative to the **majority group**
 - Then how to change the **reference category** in the regression?



1. Regressions

1.3. On categorical variables

- The function to set the reference category is **relevel()**
 - The first argument is the **vector** to indicate the reference of
 - The second argument is the value of the **reference group**

```
lm(Earnings ~ relevel(Race, "White"), asec)
```

```
## Error in relevel.default(Race, "White"): 'relevel' only for (unordered) factors
```

- Oops! I need to introduce you a **new class** of R objects first: **factors**

```
as.factor(asec$Race) %>%  
  levels()
```

```
## [1] "Black" "Other" "White"
```

```
as.factor(asec$Race) %>%  
  relevel("White") %>%  
  levels()
```

```
## [1] "White" "Black" "Other"
```



1. Regressions

1.3. On categorical variables

- The **factor class** is made for variables whose values **indicate** different **groups**
 - Values are just **arbitrary group classifiers**

```
individuals <- as.factor(c(1, 2, 3, 4, 5))
individuals[1]
```

```
## [1] 1
## Levels: 1 2 3 4 5
```

- With **factors**, R understands that the different values **do not mean anything**
 - And applying **standard operations** to factors **does not make sense**

```
individuals * 2
```

```
## Warning in Ops.factor(individuals, 2): '*' not meaningful for factors
```

```
## [1] NA NA NA NA NA
```

Practice

05:00

- 1) Open the data from the World Inequality Database we used in lecture 2
- 2) Regress the income share of the top 1% on the year variable
- 3) Redo the same regression after having converted the year variable as a **factor**

You've got 5 minutes!

Solution

1) Open the data from the World Inequality Database we used in lecture 2

```
wid <- read.csv("wid.csv")
```

2) Regress the income share of the top 1% on the year variable

```
lm(top1 ~ year, wid)
```

```
##  
## Call:  
## lm(formula = top1 ~ year, data = wid)  
##  
## Coefficients:  
## (Intercept)          year  
##  2.242e-01    -3.131e-05
```

Solution

3) Redo the same regression after having converted the year variable as a **factor**

```
lm(top1 ~ year, wid %>% mutate(year = as.factor(year)))
```

```
##  
## Call:  
## lm(formula = top1 ~ year, data = wid %>% mutate(year = as.factor(year)))  
##  
## Coefficients:  
## (Intercept)      year2011      year2012      year2013      year2014      year2015  
##   0.1621494    -0.0011366    -0.0026891    -0.0013401    -0.0004469    -0.0008683  
##   year2016      year2017      year2018      year2019  
##  -0.0001450    -0.0004857    -0.0015236    -0.0018488
```



1. Regressions

1.3. On categorical variables

- Another option to include categorical variables is to **one hot encode** the data
 - It simply means **converting** the discrete variables into dummies such that **everything is numeric**

```
asec <- asec %>% mutate(White = as.numeric(Race == "White"),  
                        Black = as.numeric(Race == "Black"),  
                        Other = as.numeric(Race == "Other"))
```

- Then we can use the **+** symbol to include n-1 categories in the regression

```
lm(Earnings ~ Black + Other, asec)
```

```
##  
## Call:  
## lm(formula = Earnings ~ Black + Other, data = asec)  
##  
## Coefficients:  
## (Intercept)      Black      Other  
##      62880      -12303       5174
```




1. Regressions

1.3. On categorical variables

- But do we really need to **omit** one **category**?

- As we are in the multivariate case let's move from $\hat{\beta} = \frac{\text{Cov}(x,y)}{\text{Var}(x)}$
- To $\hat{\beta} = (X'X)^{-1}X'y$

```
y <- as.matrix(asec$Earnings)

X <- asec %>%
  mutate(constant = 1) %>%
  select(constant, White, Black, Other) %>%
  as.matrix()
```

```
dim(X)
```

```
## [1] 64336      4
```

```
dim(t(X))
```

```
## [1]      4 64336
```

```
dim(t(X) %*% X)
```

```
## [1] 4 4
```



1. Regressions

1.3. On categorical variables

- Because of perfect **multicollinearity** it will not be possible to invert $X'X$

```
solve(t(X) %*% X)
```

```
## Error in solve.default(t(X) %*% X): system is computationally singular
```

- We have to **remove one category**

```
X <- asec %>% mutate(constant = 1) %>%  
  select(constant, Black, Other) %>%  
  as.matrix()
```

```
solve(t(X) %*% X) %*% (t(X) %*% y)
```

```
##           [,1]  
## constant 62880.488  
## Black    -12302.993  
## Other      5174.141
```

- Or to **remove the constant**

```
X <- asec %>% #mutate(constant = 1) %>%  
  select(White, Black, Other) %>%  
  as.matrix()
```

```
solve(t(X) %*% X) %*% (t(X) %*% y)
```

```
##           [,1]  
## White 62880.49  
## Black 50577.49  
## Other 68054.63
```



1. Regressions

1.3. On categorical variables

- Note that you can remove the constant in `lm()` by adding `- 1` in the formula

```
lm(Earnings ~ White + Black + Other - 1, asec)
```

```
## Coefficients:  
## White   Black   Other  
## 62880   50577   68055
```

- And that it would drop a category anyway

```
lm(Earnings ~ White + Black + Other, asec)
```

```
## Coefficients:  
## (Intercept)      White      Black      Other  
##      68055      -5174     -17477         NA
```

***But it's not because
multicollinearity does not break
lm() that you should not pay
attention to it!***



Overview

1. Regressions ✓

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables

2. Case study

- 2.1. Variable transformation
- 2.2. Functional form
- 2.3. Control variables
- 2.4. Interactions

3. Inference

- 3.1. Hypothesis testing
- 3.2. Confidence intervals

4. Report and export results

- 4.1. Regression tables
- 4.2. Plot coefficients

5. Wrap up!



Overview

1. Regressions ✓

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables

2. Case study

- 2.1. Variable transformation
- 2.2. Functional form
- 2.3. Control variables
- 2.4. Interactions



2. Case study

2.1. Variable transformation

- Imagine you want to estimate the **relationship** between **weekly hours** of work and **annual earnings**

$$\text{Earnings}_i = \alpha + \beta \text{Hours}_i + \varepsilon_i$$

- We can **estimate it in R** using the Annual Social and Economic Supplement to the US CPS

```
lm(Earnings ~ Hours, asec)
```

```
## Coefficients:  
## (Intercept)      Hours  
##      -20039      2078
```

- Are we done?

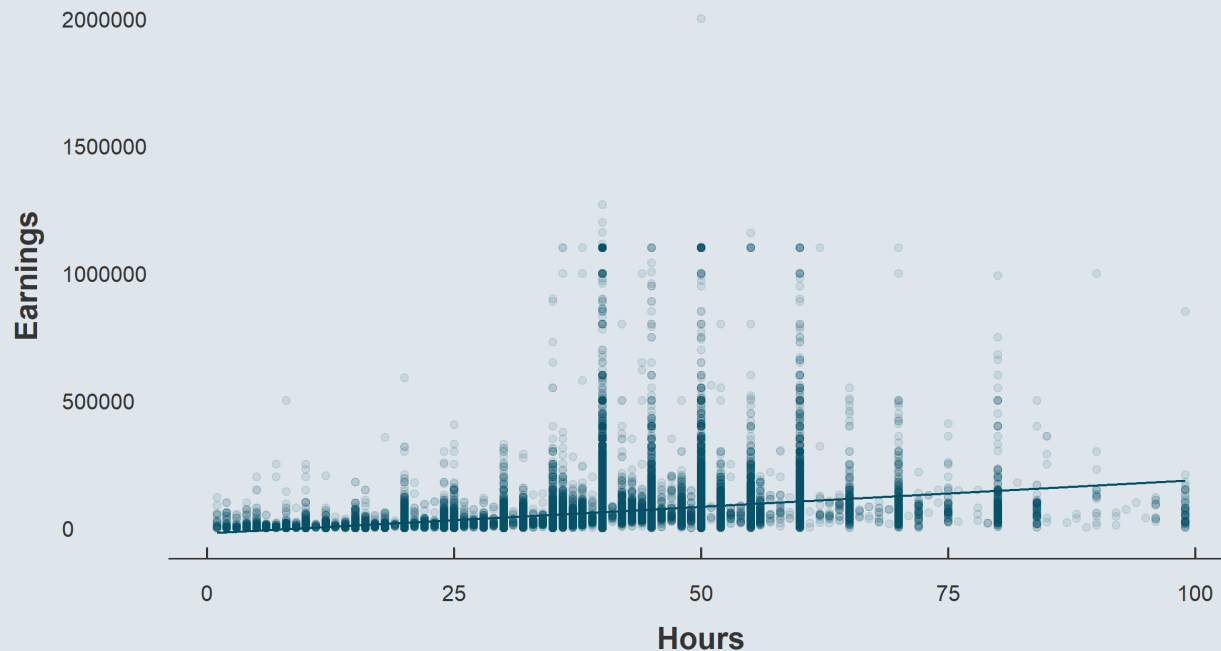
→ Let's take a look at what we just did!



2. Case study

2.1. Variable transformation

```
ggplot(asec, aes(x = Hours, y = Earnings)) +  
  geom_point() + geom_smooth(method = "lm")
```



→ Not very satisfactory

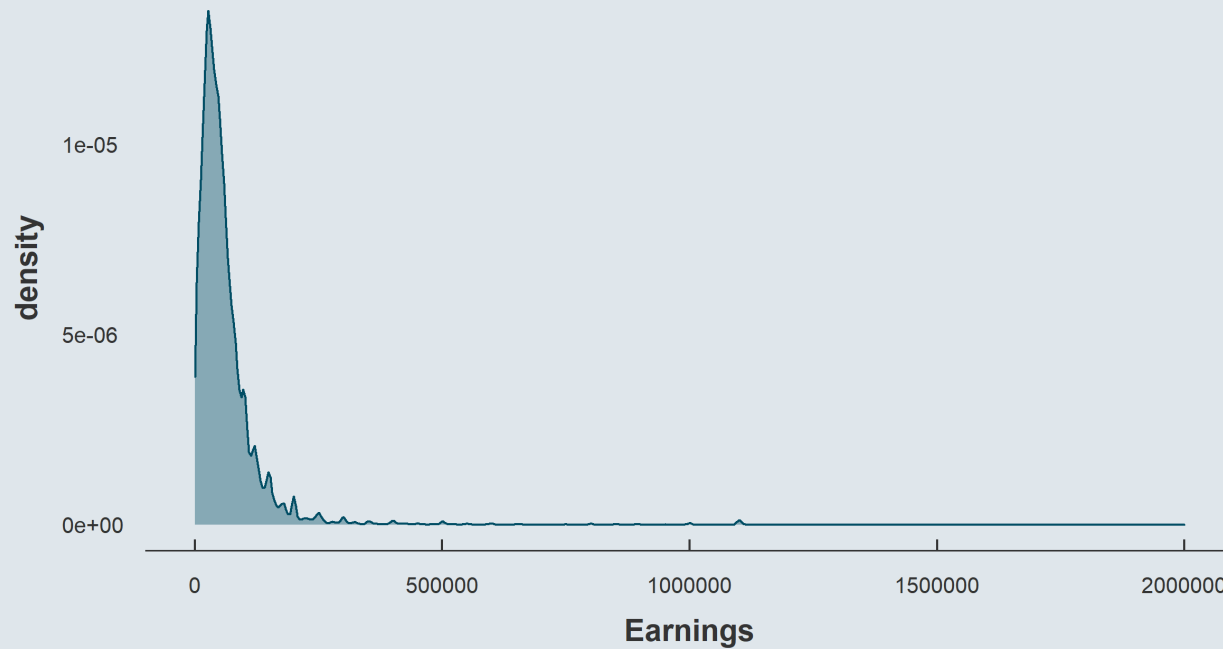
- The joint distribution does **not seem adequate** on the the **y dimension**
- Let's take a **look** at the earnings **distribution**



2. Case study

2.1. Variable transformation

```
ggplot(asec, aes(x = Earnings)) +  
  geom_density()
```



It's clearly **log-normal**

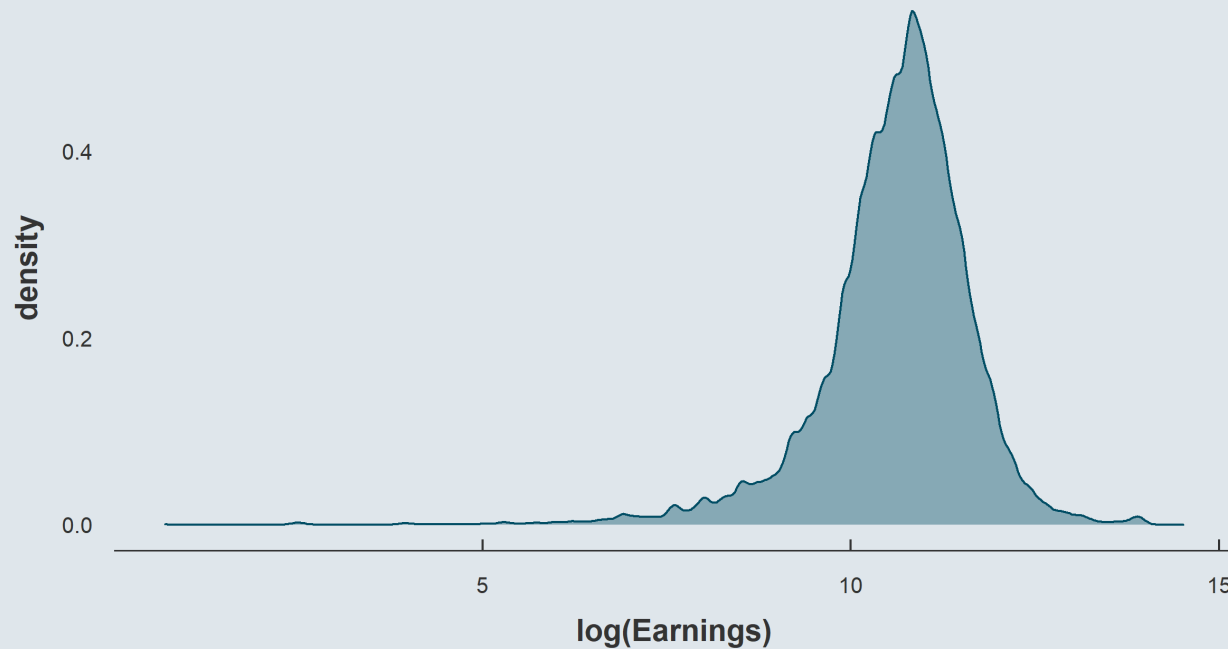
→ Let's **plot the log** of it



2. Case study

2.1. Variable transformation

```
ggplot(asec, aes(x = log(Earnings))) +  
  geom_density()
```



Better!

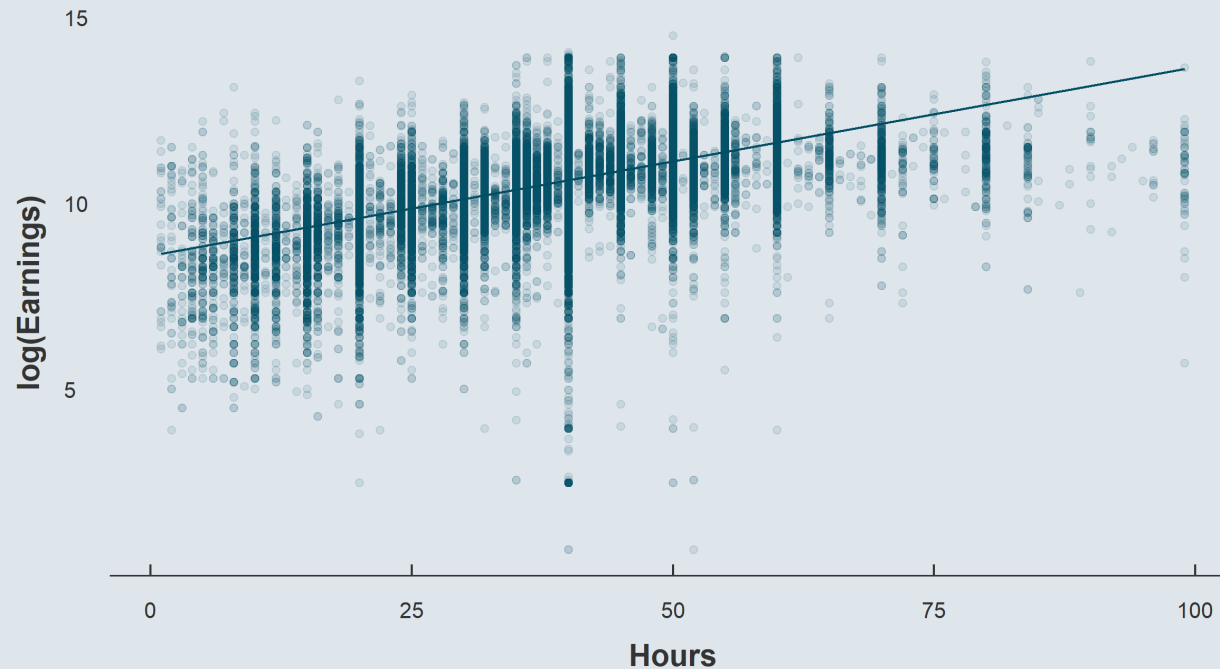
→ Let's **update** the **scatterplot**



2. Case study

2.2. Functional form

```
ggplot(asec, aes(x = Hours, y = log(Earnings))) +  
  geom_point(alpha = .1) + geom_smooth(method = "lm")
```



Definitely better!

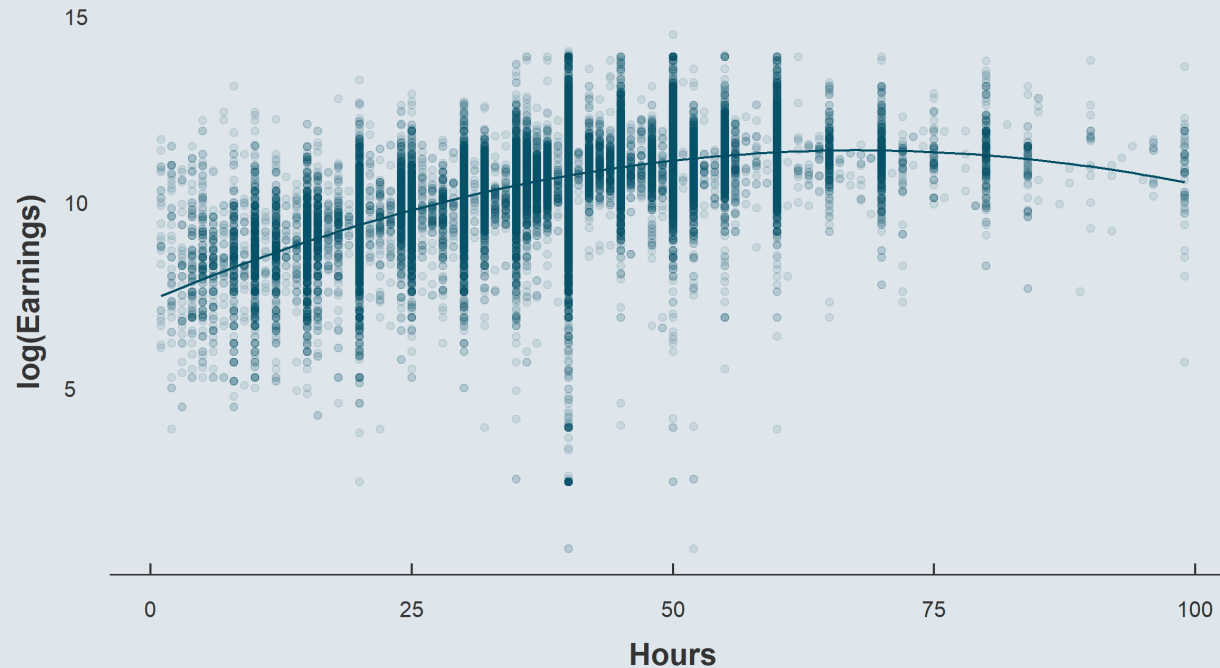
→ But something **still** feels off



2. Case study

2.2. Functional form

```
ggplot(asec, aes(x = Hours, y = log(Earnings))) +  
  geom_point(alpha = .1) + geom_smooth(method = "lm", formula = y ~ poly(x, 2))
```



→ *There we go*



2. Case study

2.2. Functional form

- We'd better rewrite our model as:

$$\log(\text{Earnings}_i) = \alpha + \beta_1 \text{Hours}_i + \beta_2 \text{Hours}_i^2 + \varepsilon_i$$

- Create the new variables

```
asec <- asec %>%  
  mutate(lEarnings = log(Earnings),  
         sqHours = Hours^2)
```

- And run the regression

```
lm(lEarnings ~ Hours + sqHours, asec)
```

```
## Coefficients:  
## (Intercept)      Hours      sqHours  
##    7.3637848    0.1192609   -0.0008804
```

Is that it?

*Isn't there something we
could/should add in our regression?*



2. Case study

2.3. Control variables

- This positive **relationship** could be **driven** by a **third variable**
 - **Males** tend both to work **part time less often** and to **earn more**
 - The **higher** the **hours**, the higher the **probability to be a male**, the higher the **expected earnings**
- Let's **control** for that in the regression

$$\log(\text{Earnings}_i) = \alpha + \beta_1 \text{Hours}_i + \beta_2 \text{Hours}_i^2 + \beta_3 \text{Male}_i + \varepsilon_i$$

- Such that Males and Females would have
 - The **same slope**
 - **Different intercepts**

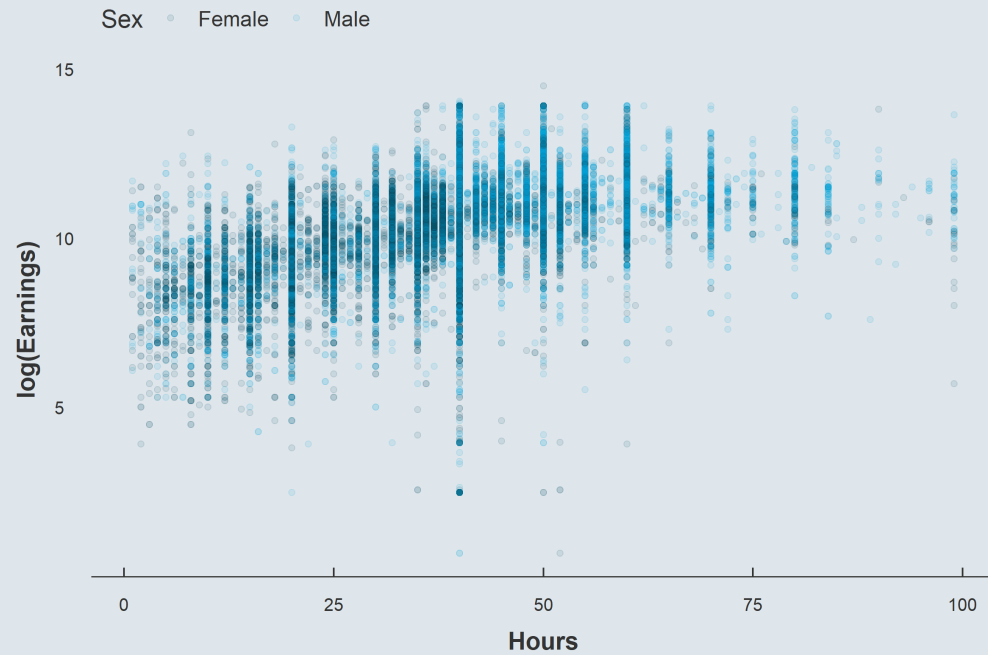
| | Male = 0 | Male = 1 |
|-----------|-----------------------------------|-----------------------------------|
| Intercept | α | $\alpha + \beta_3$ |
| Slope | $\beta_1 + 2\beta_2 \text{Hours}$ | $\beta_1 + 2\beta_2 \text{Hours}$ |

Let's take a look



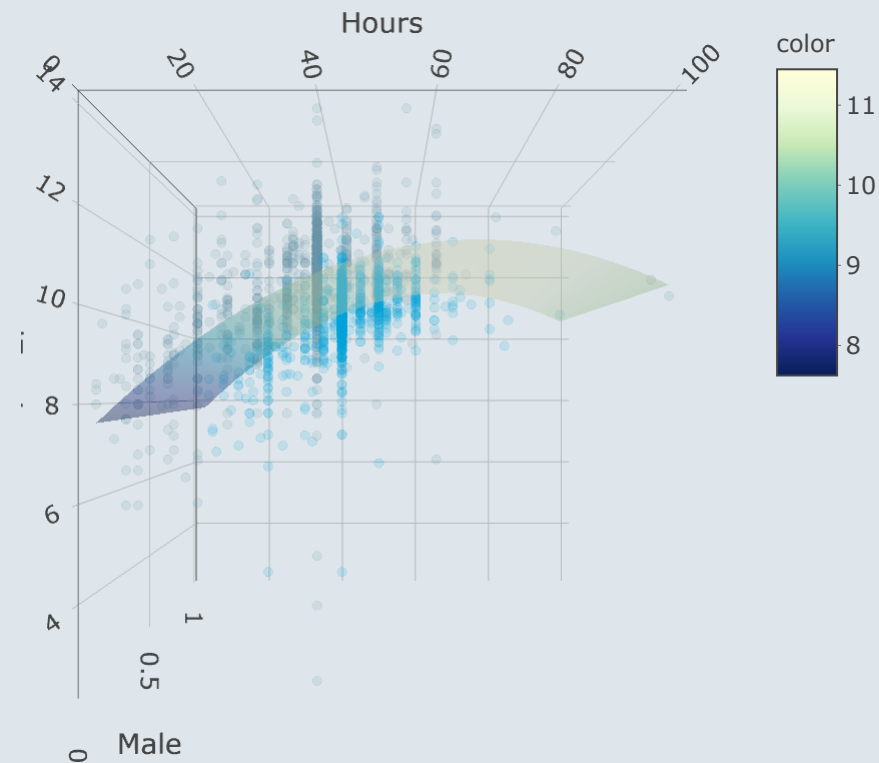
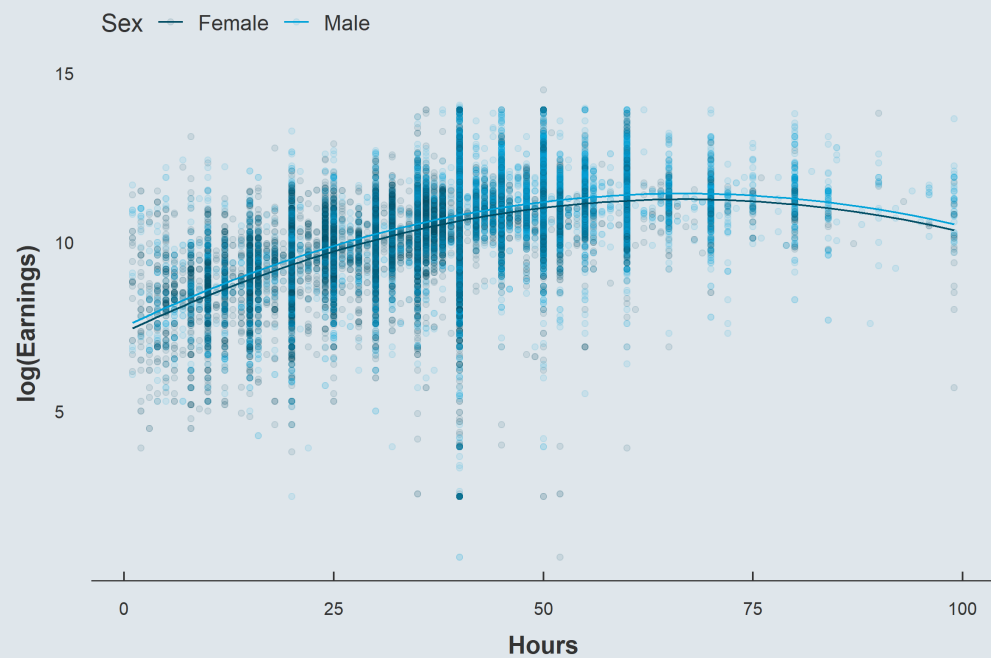
2. Case study

2.3. Control variables



2. Case study

2.3. Control variables





2. Case study

2.3. Control variables

- Once again we can include a control variable using the + symbol

```
lm(lEarnings ~ Hours + sqHours + Sex, asec)
```

```
## Coefficients:  
## (Intercept)      Hours      sqHours      SexMale  
##    7.3356057    0.1177514   -0.0008806    0.1700972
```

```
lm(lEarnings ~ Hours + sqHours, asec)
```

```
## Coefficients:  
## (Intercept)      Hours      sqHours  
##    7.3637848    0.1192609   -0.0008804
```

Actually the baseline coefficient was not that inflated

2. Case study

2.4. Interactions

- But what if we were to allow for **different slopes**?
 - The **relationship** between hours and earnings might be **heterogeneous** across sex/gender
 - This is what **interactions** allow to account for
- We simply have to include the **product** of the two variables in the model

$$\log(\text{Earnings}_i) = \alpha + \beta_1 \text{Hours}_i + \beta_2 \text{Hours}_i^2 + \beta_3 \text{Male}_i + \beta_4 \text{Hours}_i \times \text{Male}_i + \varepsilon_i$$

- Such that Males and Females would have
 - Not only **different intercepts**
 - But also **different slopes**

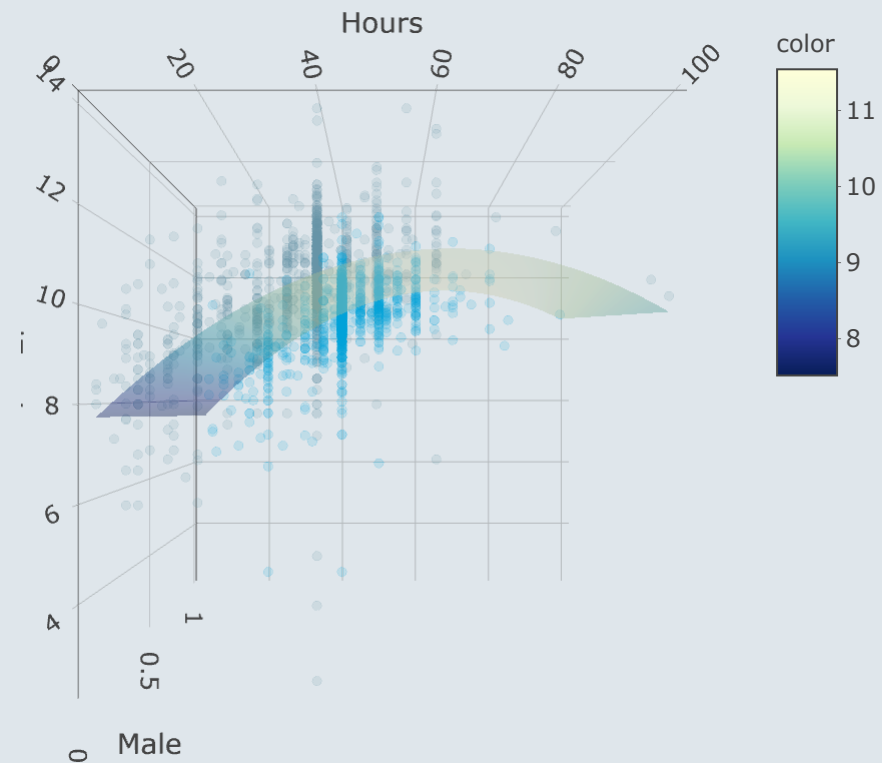
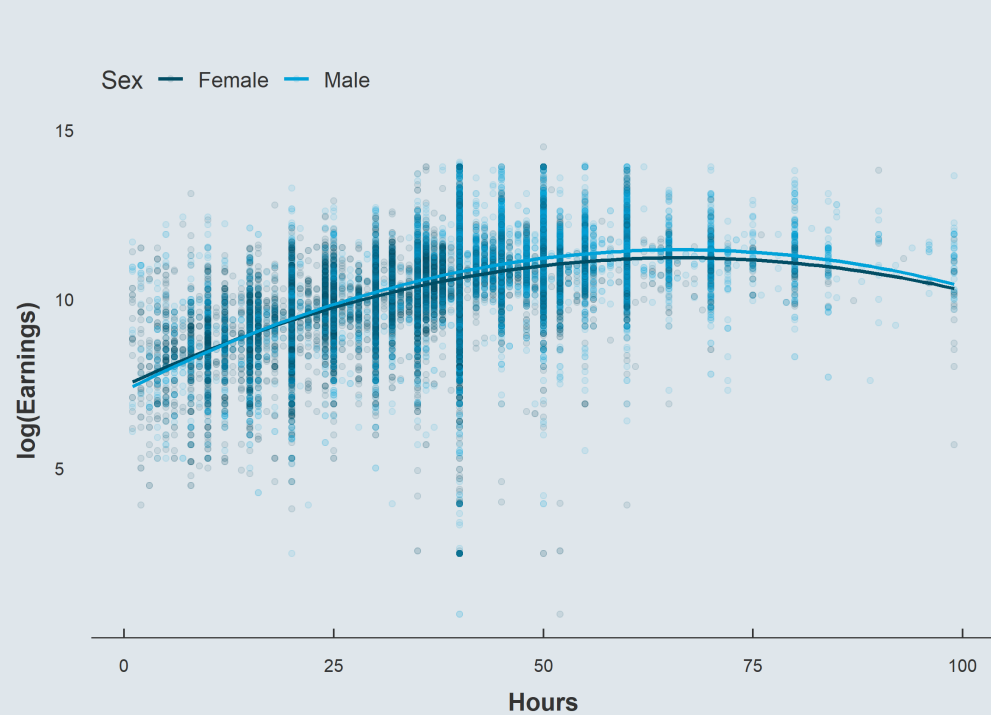
| | Male = 0 | Male = 1 |
|-----------|-----------------------------------|---|
| Intercept | α | $\alpha + \beta_3$ |
| Slope | $\beta_1 + 2\beta_2 \text{Hours}$ | $\beta_1 + 2\beta_2 \text{Hours} + \beta_4$ |

Let's take a look



2. Case study

2.4. Interactions





2. Case study

2.4. Interactions

- Now we can use the `*` symbol for products

```
results <- summary(lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec))$coefficients
results
```

| ## | | Estimate | Std. Error | t value | Pr(> t) |
|----|---------------|---------------|--------------|------------|--------------|
| ## | (Intercept) | 7.3900468002 | 2.319198e-02 | 318.646690 | 0.000000e+00 |
| ## | Hours | 0.1174998826 | 1.034522e-03 | 113.578950 | 0.000000e+00 |
| ## | sqHours | -0.0009103523 | 1.321706e-05 | -68.877048 | 0.000000e+00 |
| ## | SexMale | -0.0282905673 | 2.753222e-02 | -1.027544 | 3.041682e-01 |
| ## | Hours:SexMale | 0.0050321134 | 6.767013e-04 | 7.436240 | 1.048736e-13 |

What do you conclude ?

```
results[5, 1] / results[2, 1]
```

```
## [1] 0.04282654
```

→ A 4% difference

```
results[5, 4]
```

```
## [1] 1.048736e-13
```

→ Which is highly significant



Overview

1. Regressions ✓

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables

2. Case study ✓

- 2.1. Variable transformation
- 2.2. Functional form
- 2.3. Control variables
- 2.4. Interactions

3. Inference

- 3.1. Hypothesis testing
- 3.2. Confidence intervals

4. Report and export results

- 4.1. Regression tables
- 4.2. Plot coefficients

5. Wrap up!



Overview

1. Regressions ✓

- 1.1. On continuous variables
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3. Inference

- 3.1. Hypothesis testing
- 3.2. Confidence intervals

3. Inference

3.1. Hypothesis testing

- According to our previous regression, $\hat{\beta}_1$ is significantly **different from 0**
 - But let's pretend that you know that this coefficient is equal to **.12 in Canada**
 - How could we test whether or not our coefficient is **different from .12**?

results

| ## | | Estimate | Std. Error | t value | Pr(> t) |
|----|---------------|---------------|--------------|------------|--------------|
| ## | (Intercept) | 7.3900468002 | 2.319198e-02 | 318.646690 | 0.000000e+00 |
| ## | Hours | 0.1174998826 | 1.034522e-03 | 113.578950 | 0.000000e+00 |
| ## | sqHours | -0.0009103523 | 1.321706e-05 | -68.877048 | 0.000000e+00 |
| ## | SexMale | -0.0282905673 | 2.753222e-02 | -1.027544 | 3.041682e-01 |
| ## | Hours:SexMale | 0.0050321134 | 6.767013e-04 | 7.436240 | 1.048736e-13 |

- We can compute the t-stat from our results

$$t = \frac{\hat{\beta} - .12}{\text{s.e.}(\hat{\beta})}$$

```
t <- (results[2, 1] - .12) / results[2, 2]
t
```

```
## [1] -2.416689
```



3. Inference

3.1. Hypothesis testing

- And then we need the value of the **area outside the interval** $[-t; t]$
 - From a **Student-t** distribution with the correct number of **degrees of freedom** (#obs - #parameters)



3. Inference

3.1. Hypothesis testing

- You can get the **area below** a certain **t-value** with the **pt()** function
 - 1.
 - 2.

```
pt( , )
```




3. Inference

3.1. Hypothesis testing

- You can get the **area below** a certain **t-value** with the **pt()** function
 1. The first argument is the **t-value**
 - 2.

```
pt(t, )
```



3. Inference

3.1. Hypothesis testing

- You can get the **area below** a certain **t-value** with the **pt()** function
 1. The first argument is the **t-value**
 2. The second argument is the number of **degrees of freedom**

```
pt(t, nrow(asec) - nrow(results))
```

```
## [1] 0.007832573
```

- We just have to multiply this value by 2 to obtain the p-value

```
2 * pt(t, nrow(asec) - nrow(results))
```

```
## [1] 0.01566515
```

- Had our t-stat been positive we would have needed to multiply $1 - t$ by 2

```
2 * (1 - pt(abs(t), nrow(asec)-nrow(results)))
```

```
## [1] 0.01566515
```



3. Inference

3.1. Hypothesis testing

- A very handy function for hypothesis testing is **linearHypothesis()** from the **car** package
 -
 -

```
linearHypothesis( , )
```



3. Inference

3.1. Hypothesis testing

- A very handy function for hypothesis testing is **linearHypothesis()** from the **car** package
 - The first argument is the **model**
 -

```
linearHypothesis(lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec), )
```

3. Inference

3.1. Hypothesis testing

- A very handy function for hypothesis testing is **linearHypothesis()** from the **car** package
 - The first argument is the **model**
 - The second argument is the **hypothesis/es**

```
linearHypothesis(lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec), c("Hours = .12"))
```

```
## Linear hypothesis test
##
## Hypothesis:
## Hours = 0.12
##
## Model 1: restricted model
## Model 2: lEarnings ~ Hours + sqHours + Sex + Hours * Sex
##
##      Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    64332 46102
## 2    64331 46098   1     4.1851 5.8404 0.01567 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



3. Inference

3.1. Hypothesis testing

- It can be used for **F tests** like the one from the summary

```
linearHypothesis(lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                 c("Hours = 0", "sqHours = 0", "SexMale = 0", "Hours:SexMale = 0"))
```



3. Inference

3.1. Hypothesis testing

```
linearHypothesis(lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
  c("Hours = 0", "sqHours = 0", "SexMale = 0", "Hours:SexMale = 0"))
```

```
## Linear hypothesis test  
##  
## Hypothesis:  
## Hours = 0  
## sqHours = 0  
## SexMale = 0  
## Hours:SexMale = 0  
##  
## Model 1: restricted model  
## Model 2: lEarnings ~ Hours + sqHours + Sex + Hours * Sex  
##  
##   Res.Df  RSS Df Sum of Sq    F    Pr(>F)  
## 1   64335 68395  
## 2   64331 46098   4    22297 7779.1 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



3. Inference

```
##
## Call:
## lm(formula = lEarnings ~ Hours + sqHours + Sex + Hours * Sex,
##     data = asec)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3453  -0.4299   0.0133   0.4810   4.8506
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   7.390e+00  2.319e-02 318.647 < 2e-16 ***
## Hours         1.175e-01  1.035e-03 113.579 < 2e-16 ***
## sqHours       -9.103e-04  1.322e-05 -68.877 < 2e-16 ***
## SexMale       -2.829e-02  2.753e-02  -1.028   0.304
## Hours:SexMale  5.032e-03  6.767e-04   7.436 1.05e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8465 on 64331 degrees of freedom
## Multiple R-squared:  0.326,    Adjusted R-squared:  0.326
## F-statistic: 7779 on 4 and 64331 DF,  p-value: < 2.2e-16
```


3. Inference

3.2. Confidence intervals

- Now we know how to get the area below a given value of a Student t distribution
 - But sometimes the opposite is useful as well
 - In particular to compute **confidence intervals**
- Indeed the confidence interval for a $\hat{\beta}$ coefficient is given by:

$$\hat{\beta} \pm t_{1-(\alpha/2), \text{df}} \times \text{s.e.}(\hat{\beta})$$

- Where α denotes the desired significance level
- **qt()** gives the t -statistic below which lies the desired share of the area of a given Student t distribution
 -
 -

```
qt( , )
```

3. Inference

3.2. Confidence intervals

- Now we know how to get the area below a given value of a Student t distribution
 - But sometimes the opposite is useful as well
 - In particular to compute **confidence intervals**
- Indeed the confidence interval for a $\hat{\beta}$ coefficient is given by:

$$\hat{\beta} \pm t_{1-(\alpha/2), \text{df}} \times \text{s.e.}(\hat{\beta})$$

- Where α denotes the desired significance level
- **qt()** gives the t -statistic below which lies the desired share of the area of a given Student t distribution
 - The first argument is $1 - (\alpha/2)$
 -

```
qt(.975, )
```

3. Inference

3.2. Confidence intervals

- Now we know how to get the area below a given value of a Student t distribution
 - But sometimes the opposite is useful as well
 - In particular to compute **confidence intervals**
- Indeed the confidence interval for a $\hat{\beta}$ coefficient is given by:

$$\hat{\beta} \pm t_{1-(\alpha/2), \text{df}} \times \text{s.e.}(\hat{\beta})$$

- Where α denotes the desired significance level
- **qt()** gives the t -statistic below which lies the desired share of the area of a given Student t distribution
 - The first argument is $1 - (\alpha/2)$
 - The second argument is the number of **degrees of freedom**

```
qt(.975, Inf)
```

```
## [1] 1.959964
```



3. Inference

3.2. Confidence intervals

- So if we want a 97% **confidence interval** for our coefficients associated with hours, we feed **qt()** with:
 -
 -

```
t003 <- qt( , )
```



3. Inference

3.2. Confidence intervals

- So if we want a 97% **confidence interval** for our coefficients associated with hours, we feed **qt()** with:
 - The **share** of the Student t **distribution** below the desired t -stat: $1 - (\alpha/2) = 1 - (0.03/2) = 0.985$
 -

```
t003 <- qt(.985, )
```

3. Inference

3.2. Confidence intervals

- So if we want a 97% **confidence interval** for our coefficients associated with hours, we feed **qt()** with:
 - The **share** of the Student t **distribution** below the desired t -stat: $1 - (\alpha/2) = 1 - (0.03/2) = 0.985$
 - The **degrees of freedom**: $\#obs. - \#params$

```
t003 <- qt(.985, nrow(asec) - nrow(results))
```



3. Inference

3.2. Confidence intervals

- So if we want a 97% **confidence interval** for our coefficients associated with hours, we feed **qt()** with:
 - The **share** of the Student t **distribution** below the desired t -stat: $1 - (\alpha/2) = 1 - (0.03/2) = 0.985$
 - The **degrees of freedom**: $\#obs. - \#params$

```
t003 <- qt(.985, nrow(asec) - nrow(results))  
t003
```

```
## [1] 2.170139
```

- And we apply the formula:

```
results[2, 1] - (results[2, 2] * t003)
```

```
## [1] 0.1152548
```

```
results[2, 1] + (results[2, 2] * t003)
```

```
## [1] 0.1197449
```



Overview

1. Regressions ✓

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables

2. Case study ✓

- 2.1. Variable transformation
- 2.2. Functional form
- 2.3. Control variables
- 2.4. Interactions

3. Inference ✓

- 3.1. Hypothesis testing
- 3.2. Confidence intervals

4. Report and export results

- 4.1. Regression tables
- 4.2. Plot coefficients

5. Wrap up!



Overview

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4. Report and export results

4.1. Regression tables

- The output of the **summary()** function is very **practical** but **not** very **convenient** to report the main results
 - Academic regression tables** rather look like that

Table 5

The effect of distance to the line on production levels.

| | Dependent Variable Ln Per Capita GDP | | | | | |
|---|--------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Ln Distance to Historical Lines | -0.0617
(0.0286) | -0.0434
(0.0281) | -0.0491
(0.0277) | -0.0581
(0.0265) | -0.0699
(0.0270) | -0.0681
(0.0272) |
| Ln Distance to Segment City | | 0.065
(0.232) | -0.061
(0.266) | -0.063
(0.282) | -0.178
(0.325) | -0.208
(0.298) |
| Ln Distance to Segment City ² | | -0.0276
(0.0277) | -0.0089
(0.0308) | -0.0071
(0.0324) | 0.0072
(0.0372) | 0.0101
(0.0344) |
| Ln Distance to Navigable River | | | 0.318
(0.153) | 0.321
(0.141) | 0.366
(0.140) | 0.385
(0.138) |
| Ln Distance to Navigable River ² | | | -0.0517
(0.0200) | -0.0481
(0.0186) | -0.0550
(0.0188) | -0.0554
(0.0183) |
| Ln Area | | | | -1.572
(0.681) | -1.442
(0.635) | -1.441
(0.642) |
| Ln Area ² | | | | 0.0983
(0.0486) | 0.0911
(0.0459) | 0.0894
(0.0464) |
| Ln Distance to Coastline | | | | | -0.243
(0.224) | -0.207
(0.219) |
| Ln Distance to Coastline ² | | | | | 0.0168
(0.0265) | 0.0141
(0.0259) |
| Ln Distance to Country Border | | | | | | -16.49
(6.097) |
| Ln Distance to Country Border ² | | | | | | 1.241
(0.459) |
| Observations | 2744 | 2744 | 2744 | 2744 | 2744 | 2744 |
| R-squared | 0.818 | 0.826 | 0.833 | 0.845 | 0.849 | 0.852 |

Notes: All regressions control for year and province fixed effects. Standard errors are clustered at the county level. These estimates use an unbalanced county-year level panel. GDP data are from Provincial Statistical Yearbooks. All geographic variables are computed by the authors.



4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables.
 -
 -
 -
 -
 -
 -

```
outreg <- huxreg()  
#  
#  
#  
#  
#  
#  
#  
#
```



4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables. Main arguments include:
 - As many **models** as you want, named or not
 -
 -
 -
 -
 -

```
outreg <- huxreg(Baseline = lm(lEarnings ~ Hours, asec),  
                 lm(lEarnings ~ Hours + sqHours, asec),  
                 lm(lEarnings ~ Hours + sqHours + Sex, asec),  
                 lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec))
```

```
#  
#  
#  
#  
#
```



4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables. Main arguments include:
 - As many **models** as you want, named or not
 - Which **uncertainty statistic** to display (std.error, p.value, conf.low, conf.high)
 -
 -
 -
 -

```
outreg <- huxreg(Baseline = lm(lEarnings ~ Hours, asec),  
                 lm(lEarnings ~ Hours + sqHours, asec),  
                 lm(lEarnings ~ Hours + sqHours + Sex, asec),  
                 lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                 error_format = "{std.error}")
```

```
#  
#  
#  
#
```



4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables. Main arguments include:
 - As many **models** as you want, named or not
 - Which **uncertainty statistic** to display (std.error, p.value, conf.low, conf.high)
 - Where to **place** the uncertainty statistic (below, same, right)
 -
 -
 -

```
outreg <- huxreg(Baseline = lm(lEarnings ~ Hours, asec),  
                lm(lEarnings ~ Hours + sqHours, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                error_format = "{std.error}",  
                error_pos = "below")
```

```
#  
#  
#
```



4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables. Main arguments include:
 - As many **models** as you want, named or not
 - Which **uncertainty statistic** to display (std.error, p.value, conf.low, conf.high)
 - Where to **place** the uncertainty statistic (below, same, right)
 - Which **general statistics** to display (adj.r.squared, df, ...)
 -
 -

```
outreg <- huxreg(Baseline = lm(lEarnings ~ Hours, asec),  
                lm(lEarnings ~ Hours + sqHours, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                error_format = "{std.error}",  
                error_pos = "below",  
                statistics = c(N = "nobs", R2 = "r.squared"))  
  
#  
#
```

4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables. Main arguments include:
 - As many **models** as you want, named or not
 - Which **uncertainty statistic** to display (std.error, p.value, conf.low, conf.high)
 - Where to **place** the uncertainty statistic (below, same, right)
 - Which **general statistics** to display (adj.r.squared, df, ...)
 - The desired **significance symbology**
 -

```
outreg <- huxreg(Baseline = lm(lEarnings ~ Hours, asec),  
                lm(lEarnings ~ Hours + sqHours, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                error_format = "{std.error}",  
                error_pos = "below",  
                statistics = c(N = "nobs", R2 = "r.squared"),  
                stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1))
```

#

4. Report and export results

4.1. Regression tables

- The **huxreg()** function from **huxtable** allows to create such tables. Main arguments include:
 - As many **models** as you want, named or not
 - Which **uncertainty statistic** to display (std.error, p.value, conf.low, conf.high)
 - Where to **place** the uncertainty statistic (below, same, right)
 - Which **general statistics** to display (adj.r.squared, df, ...)
 - The desired **significance symbology**
 - What to write in the **table footnote**

```
outreg <- huxreg(Baseline = lm(lEarnings ~ Hours, asec),  
                lm(lEarnings ~ Hours + sqHours, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex, asec),  
                lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                error_format = "{std.error}",  
                error_pos = "below",  
                statistics = c(N = "nobs", R2 = "r.squared"),  
                stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),  
                note = "Dependent variable: log annual earnings. {stars}")
```



4. Report and export results

4.1. Regression tables

| | Baseline | (2) | (3) | (4) |
|---------------|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| sqHours | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| SexMale | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| Hours:SexMale | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |

Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1

- Then export it with `quick_[latex/html/pdf/docx](outreg, file = "path/filename.format")`



4. Report and export results

4.1. Regression tables

```
1 \documentclass{article}
2 \usepackage{array}
3 \usepackage{caption}
4 \usepackage{graphicx}
5 \usepackage{siunitx}
6 \usepackage[normalem]{ulem}
7 \usepackage{colortbl}
8 \usepackage{multirow}
9 \usepackage{hhline}
10 \usepackage{calc}
11 \usepackage{tabularx}
12 \usepackage{threeparttable}
13 \usepackage{wrapfig}
14 \usepackage{adjustbox}
15 \usepackage{hyperref}
16 % These are LaTeX packages. You can install them using your LaTeX management
17 % software,
18 % or by running 'huxtable::install_latex_dependencies()' from within R.
19 % Other packages may be required if you use non-standard tabulars (e.g. tabularray).
20 \pagenumbering{gobble}
21 \begin{document}
22
23
24 \providecommand{\huxb}[2]{\arrayrulecolor[RGB]{#1}\global\arrayrulewidth=#2pt}
25 \providecommand{\huxvb}[2]{\color[RGB]{#1}\vrule width #2pt}
26 \providecommand{\huxtpad}[1]{\rule{0pt}{#1}}
27 \providecommand{\huxbpad}[1]{\rule[-#1]{0pt}{#1}}
28
29 \begin{table}[ht]
30 \begin{centerbox}
31 \begin{threeparttable}
32 \setlength{\tabcolsep}{0pt}
33 \begin{tabular}{l l l l l}
34
35
36 \hhline>{\huxb{0, 0, 0}{0.8}}->{\huxb{0, 0, 0}{0.8}}->{\huxb{0, 0, 0}{0.8}}->
37 {\huxb{0, 0, 0}{0.8}}->{\huxb{0, 0, 0}{0.8}}-
38 \arrayrulecolor{black}
```

| | Baseline | (2) | (3) | (4) |
|---------------|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| sqHours | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| SexMale | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| Hours:SexMale | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |

Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1



4. Report and export results

4.1. Regression tables

```
1 <!DOCTYPE html>
2 <html lang="French-France">
3 <head><meta charset="utf8"><title>hux.html</title></head>
4 <body>
5
6 <p>&nbsp;</p><table class="huxtable" style="border-collapse: collapse; border: 0px; margin-bottom: 2em
7 <col><col><col><col><tr>
8 <th style="vertical-align: top; text-align: center; white-space: normal; border-style: solid solid sol
9 <tr>
10 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
11 <tr>
12 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
13 <tr>
14 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
15 <tr>
16 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
17 <tr>
18 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
19 <tr>
20 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
21 <tr>
22 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
23 <tr>
24 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
25 <tr>
26 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
27 <tr>
28 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
29 <tr>
30 <th style="vertical-align: top; text-align: left; white-space: normal; padding: 6pt 6pt 6pt 6pt; font-f
31 <tr>
32 <th style="vertical-align: top; text-align: left; white-space: normal; border-style: solid solid solid
33 <tr>
34 <th colspan="5" style="vertical-align: top; text-align: left; white-space: normal; border-style: solid
35 </table>
36
37 </body></html>
```

| | Baseline | (2) | (3) | (4) |
|---------------|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| sqHours | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| SexMale | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| Hours:SexMale | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |

Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1



| | Baseline | (2) | (3) | (4) |
|---|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| sqHours | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| SexMale | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| Hours:SexMale | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |
| Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1 | | | | |



4. Report and export results

4.1. Regression tables

| | Baseline | (2) | (3) | (4) |
|---------------|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| sqHours | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| SexMale | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| Hours:SexMale | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |

Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1

| | Baseline | (2) | (3) | (4) |
|----------------------|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| <u>sqHours</u> | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| <u>SexMale</u> | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| <u>Hours:SexMale</u> | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |

Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1



Practice

10:00

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce this table:

| | | Dependent variable: Log annual earnings | | | |
|----|--|---|------------|------------|------------|
| | | (1) | (2) | (3) | (4) |
| | | <hr/> | | | |
| ## | Hours worked | 0.051 *** | 0.119 *** | 0.118 *** | 0.117 *** |
| ## | | (0.000) | (0.000) | (0.000) | (0.000) |
| ## | (Hours worked) ² | | -0.001 *** | -0.001 *** | -0.001 *** |
| ## | | | (0.000) | (0.000) | (0.000) |
| ## | Male | | | 0.170 *** | -0.028 |
| ## | | | | (0.000) | (0.304) |
| ## | Hours worked x Male | | | | 0.005 *** |
| ## | | | | | (0.000) |
| ## | Constant | 8.604 *** | 7.364 *** | 7.336 *** | 7.390 *** |
| ## | | (0.000) | (0.000) | (0.000) | (0.000) |
| ## | | <hr/> | | | |
| ## | N | 64336 | 64336 | 64336 | 64336 |
| ## | R2 | 0.268 | 0.319 | 0.325 | 0.326 |
| ## | <hr/> | | | | |
| ## | Significance: *** p < 0.01; ** p < 0.05; * p < 0.1 | | | | |

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),  
      lm(lEarnings ~ Hours + sqHours, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec))  
#  
#  
#  
#  
#  
#  
#  
#  
#  
#  
#  
#  
#
```


Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),  
      lm(lEarnings ~ Hours + sqHours, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
      error_format = "{p.value}",  
      error_pos = "below")
```

```
#  
#  
#  
#  
#  
#  
#  
#  
#  
#  
#
```

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),  
      lm(lEarnings ~ Hours + sqHours, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
      error_format = "{p.value}",  
      error_pos = "below",  
      statistics = c(N = "nobs", R2 = "r.squared"))
```

```
#  
#  
#  
#  
#  
#  
#  
#  
#  
#
```

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),  
      lm(lEarnings ~ Hours + sqHours, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
      error_format = "{p.value}",  
      error_pos = "below",  
      statistics = c(N = "nobs", R2 = "r.squared"),  
      stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),  
      note = "Significance: {stars}")
```

```
#  
#  
#  
#  
#  
#  
#  
#
```

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),  
      lm(lEarnings ~ Hours + sqHours, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex, asec),  
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
      error_format = "{p.value}",  
      error_pos = "below",  
      statistics = c(N = "nobs", R2 = "r.squared"),  
      stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),  
      note = "Significance: {stars}",  
      align = "c")
```

```
#  
#  
#  
#  
#  
#  
#
```

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),
      lm(lEarnings ~ Hours + sqHours, asec),
      lm(lEarnings ~ Hours + sqHours + Sex, asec),
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),
      error_format = "{p.value}",
      error_pos = "below",
      statistics = c(N = "nobs", R2 = "r.squared"),
      stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
      note = "Significance: {stars}",
      align = "c",
      coefs = c("Hours worked" = "Hours",
                "(Hours worked)2" = "sqHours",
                "Male" = "SexMale",
                "Hours worked x Male" = "Hours:SexMale",
                "Constant" = "(Intercept)"))
#
#
```

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),
      lm(lEarnings ~ Hours + sqHours, asec),
      lm(lEarnings ~ Hours + sqHours + Sex, asec),
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),
      error_format = "{p.value}",
      error_pos = "below",
      statistics = c(N = "nobs", R2 = "r.squared"),
      stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
      note = "Significance: {stars}",
      align = "c",
      coefs = c("Hours worked" = "Hours",
                "(Hours worked)2" = "sqHours",
                "Male" = "SexMale",
                "Hours worked x Male" = "Hours:SexMale",
                "Constant" = "(Intercept)")) %>%
insert_row(c("", rep("Dependent variable: Log annual earnings", 4)), after = 0)
```

#

Solution

Use the functions `huxreg()`, `insert_row()`, and `merge_cells()` to reproduce the table

```
huxreg(lm(lEarnings ~ Hours, asec),
      lm(lEarnings ~ Hours + sqHours, asec),
      lm(lEarnings ~ Hours + sqHours + Sex, asec),
      lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),
      error_format = "{p.value}",
      error_pos = "below",
      statistics = c(N = "nobs", R2 = "r.squared"),
      stars = c(`***` = 0.01, `**` = 0.05, `*` = 0.1),
      note = "Significance: {stars}",
      align = "c",
      coefs = c("Hours worked" = "Hours",
                "(Hours worked)2" = "sqHours",
                "Male" = "SexMale",
                "Hours worked x Male" = "Hours:SexMale",
                "Constant" = "(Intercept)")) %>%
insert_row(c("", rep("Dependent variable: Log annual earnings", 4)), after = 0) %>%
merge_cells(1, 2:5)
```



4. Report and export results

4.2. Plot coefficients

- It can also be useful to provide a **graphical representation** of the coefficients
 - By now you should be able to work it around with **dplyr** and **ggplot**
 - But there exists a **shortcut**
- The **plot_summs()** function from the **jtools** package takes regression models as inputs and **plots the results**
 -
 -
 -
 -
 -

```
plot_summs()
```

```
#
```

```
#
```

```
#
```

```
#
```

```
#
```


4. Report and export results

4.2. Plot coefficients

- It can also be useful to provide a **graphical representation** of the coefficients
 - By now you should be able to work it around with **dplyr** and **ggplot**
 - But there exists a **shortcut**
- The **plot_summs()** function from the **jtools** package takes regression models as inputs and **plots the results**
 - First feed it with your **models**
 -
 -
 -
 -

```
plot_summs(lm(lEarnings ~ Hours + Race + Sex, asec),
           lm(lEarnings ~ Hours + Race + Sex + sqHours, asec))
```

```
#
#
#
#
```



4. Report and export results

4.2. Plot coefficients

- It can also be useful to provide a **graphical representation** of the coefficients
 - By now you should be able to work it around with **dplyr** and **ggplot**
 - But there exists a **shortcut**
- The **plot_summs()** function from the **jtools** package takes regression models as inputs and **plots the results**
 - First feed it with your **models**
 - You can choose to **omit** some **coefficients**
 -
 -
 -

```
plot_summs(lm(lEarnings ~ Hours + Race + Sex, asec),  
           lm(lEarnings ~ Hours + Race + Sex + sqHours, asec),  
           omit.coefs = "(Intercept)")
```

```
#
```

```
#
```

```
#
```

4. Report and export results

4.2. Plot coefficients

- It can also be useful to provide a **graphical representation** of the coefficients
 - By now you should be able to work it around with **dplyr** and **ggplot**
 - But there exists a **shortcut**
- The **plot_summs()** function from the **jtools** package takes regression models as inputs and **plots the results**
 - First feed it with your **models**
 - You can choose to **omit** some **coefficients**
 - Change the **level** of the **confidence** intervals
 -
 -

```
plot_summs(lm(lEarnings ~ Hours + Race + Sex, asec),
           lm(lEarnings ~ Hours + Race + Sex + sqHours, asec),
           omit.coefs = "(Intercept)",
           ci_level = 0.99)
```

```
#
```

```
#
```

4. Report and export results

4.2. Plot coefficients

- It can also be useful to provide a **graphical representation** of the coefficients
 - By now you should be able to work it around with **dplyr** and **ggplot**
 - But there exists a **shortcut**
- The **plot_summs()** function from the **jtools** package takes regression models as inputs and **plots the results**
 - First feed it with your **models**
 - You can choose to **omit** some **coefficients**
 - Change the **level** of the **confidence** intervals
 - Custom the **color palette**
 -

```
plot_summs(lm(lEarnings ~ Hours + Race + Sex, asec),
           lm(lEarnings ~ Hours + Race + Sex + sqHours, asec),
           omit.coefs = "(Intercept)",
           ci_level = 0.99,
           colors = c("#014D64", "#00A2D9"))
```

#

4. Report and export results

4.2. Plot coefficients

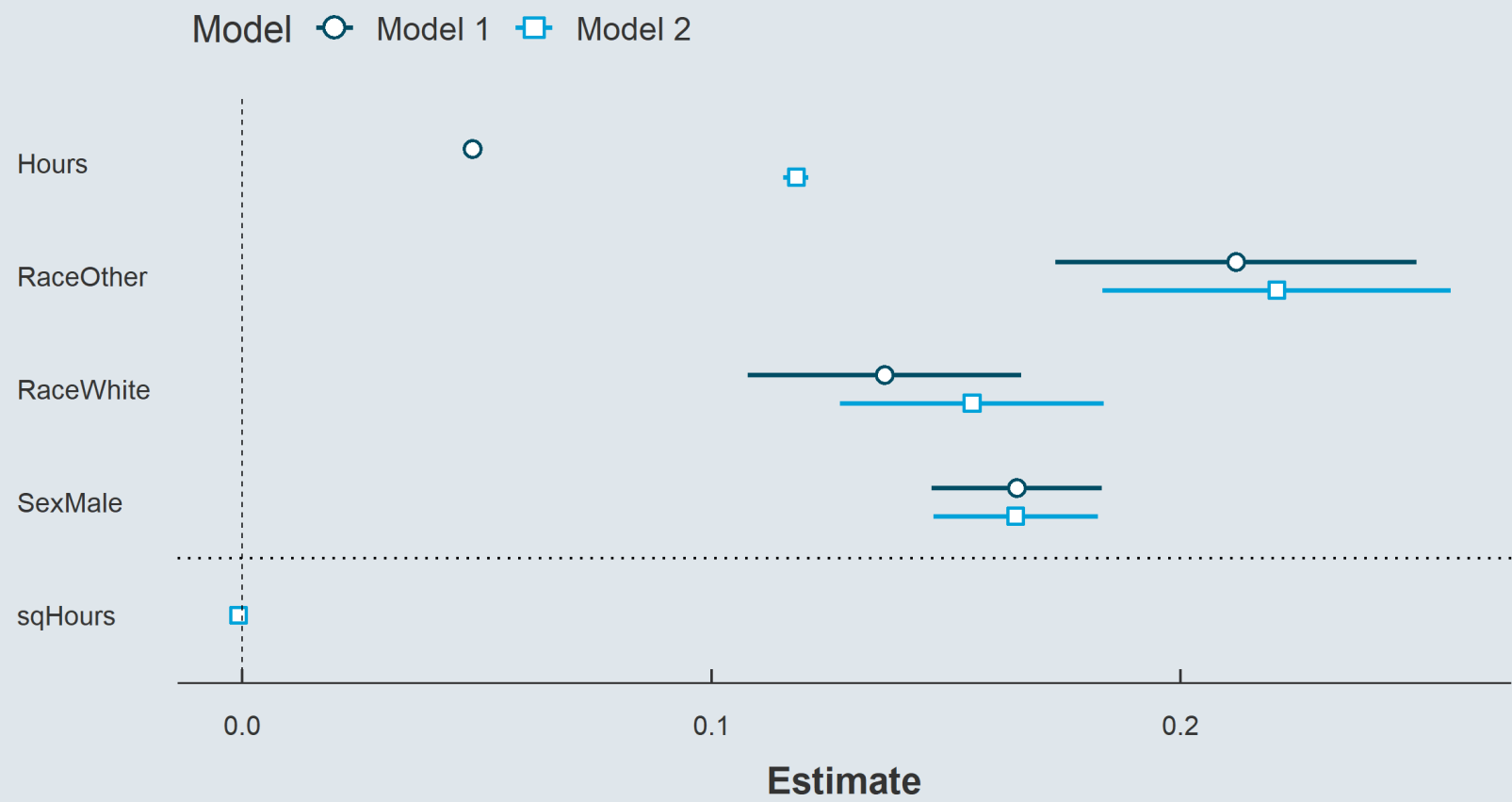
- It can also be useful to provide a **graphical representation** of the coefficients
 - By now you should be able to work it around with **dplyr** and **ggplot**
 - But there exists a **shortcut**
- The **plot_summs()** function from the **jtools** package takes regression models as inputs and **plots the results**
 - First feed it with your **models**
 - You can choose to **omit** some **coefficients**
 - Change the **level** of the **confidence** intervals
 - Custom the **color palette**
 - And add ggplot functions!

```
plot_summs(lm(lEarnings ~ Hours + Race + Sex, asec),
           lm(lEarnings ~ Hours + Race + Sex + sqHours, asec),
           omit.coefs = "(Intercept)",
           ci_level = 0.99,
           colors = c("#014D64", "#00A2D9")) +
  geom_hline(yintercept = 1.5, linetype = "dotted")
```



4. Report and export results

4.2. Plot coefficients





Overview

1. Regressions ✓

- 1.1. On continuous variables
- 1.2. On binary variables
- 1.3. On categorical variables

2. Case study ✓

- 2.1. Variable transformation
- 2.2. Functional form
- 2.3. Control variables
- 2.4. Interactions

3. Inference ✓

- 3.1. Hypothesis testing
- 3.2. Confidence intervals

4. Report and export results ✓

- 4.1. Regression tables
- 4.2. Plot coefficients

5. Wrap up!



5. Wrap up!

Regressions in R

```
summary(lm(formula = ige ~ gini, data = ggcurve))
```

```
##
## Call:
## lm(formula = ige ~ gini, data = ggcurve)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.188991 -0.088238 -0.000855  0.047284  0.252310
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09129     0.12870  -0.709   0.48631
## gini         1.01546     0.26425   3.843   0.00102 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1159 on 20 degrees of freedom
## Multiple R-squared:  0.4247,    Adjusted R-squared:  0.396
## F-statistic: 14.77 on 1 and 20 DF,  p-value: 0.001016
```

← Command

← Residuals distribution

← Coefs, s.e., t-/p-values

← Significance

← Residual s.e. & df.

← R^2 & adjusted R^2

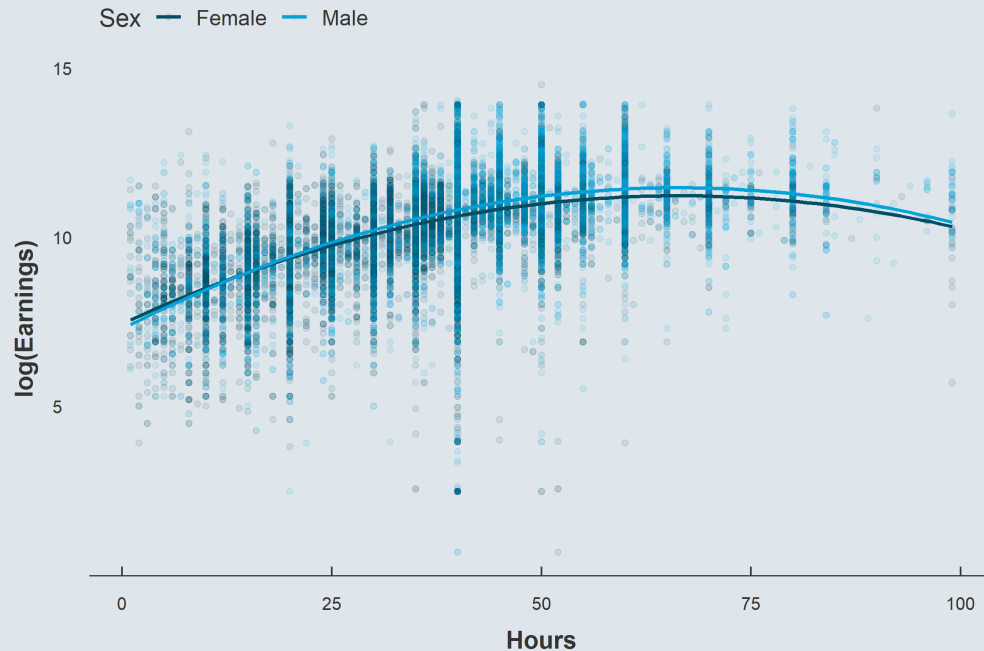
← F-test results



5. Wrap up!

Variable transformations, functional forms, controls, interactions

$$\log(\text{Earnings}_i) = \alpha + \beta_1 \text{Hours}_i + \beta_2 \text{Hours}_i^2 + \beta_3 \text{Male}_i + \beta_4 \text{Hours}_i \times \text{Male}_i + \varepsilon_i$$



```
summary(  
  lm(lEarnings ~ Hours +  
    sqHours +  
    Sex +  
    Hours * Sex,  
    asec)  
)$coefficients[, 1:2]
```

| ## | Estimate | Std. Error |
|------------------|---------------|--------------|
| ## (Intercept) | 7.3900468002 | 2.319198e-02 |
| ## Hours | 0.1174998826 | 1.034522e-03 |
| ## sqHours | -0.0009103523 | 1.321706e-05 |
| ## SexMale | -0.0282905673 | 2.753222e-02 |
| ## Hours:SexMale | 0.0050321134 | 6.767013e-04 |



5. Wrap up!

Hypothesis testing

```
linearHypothesis(lm(lEarnings ~ Hours + sqHours + Sex + Hours * Sex, asec),  
                  c("Hours = 0", "sqHours = 0", "SexMale = 0", "Hours:SexMale = 0"))
```

```
## Linear hypothesis test  
##  
## Hypothesis:  
## Hours = 0  
## sqHours = 0  
## SexMale = 0  
## Hours:SexMale = 0  
##  
## Model 1: restricted model  
## Model 2: lEarnings ~ Hours + sqHours + Sex + Hours * Sex  
##  
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)  
## 1   64335 68395  
## 2   64331 46098   4    22297 7779.1 < 2.2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



5. Wrap up!

Report/export results

```
huxreg(  
  Baseline = lm(lEarnings ~ Hours, asec),  
  lm(lEarnings ~ Hours + sqHours, asec),  
  lm(lEarnings ~ Hours + sqHours + Sex, asec),  
  lm(lEarnings ~ Hours + sqHours + Sex +  
    Hours * Sex, asec),  
  error_format = "{std.error}",  
  error_pos = "below",  
  statistics = c(N="nobs", R2="r.squared"),  
  stars = c(`***` = 0.01,  
    `**` = 0.05,  
    `*` = 0.1),  
  note = paste("Dependent variable: log",  
    "annual earnings. {stars}")  
)
```

| | Baseline | (2) | (3) | (4) |
|---------------|----------------------|-----------------------|-----------------------|-----------------------|
| (Intercept) | 8.604 ***
(0.014) | 7.364 ***
(0.022) | 7.336 ***
(0.022) | 7.390 ***
(0.023) |
| Hours | 0.051 ***
(0.000) | 0.119 ***
(0.001) | 0.118 ***
(0.001) | 0.117 ***
(0.001) |
| sqHours | | -0.001 ***
(0.000) | -0.001 ***
(0.000) | -0.001 ***
(0.000) |
| SexMale | | | 0.170 ***
(0.007) | -0.028
(0.028) |
| Hours:SexMale | | | | 0.005 ***
(0.001) |
| N | 64336 | 64336 | 64336 | 64336 |
| R2 | 0.268 | 0.319 | 0.325 | 0.326 |

Dependent variable: log annual earnings. *** p < 0.01; ** p < 0.05; * p < 0.1